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Analyzing the carbon footprint and energy usage of producing alternative nitrogen fertilizers in the Swedish context

– A factor-based scenario analysis conducted comparing Swedish and European urea production

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Abstract

Sweden aims to become independent in nitrogen fertilizer production after the international market proved volatile under the Russian invasion of Ukraine. Domestic organic recycling of nitrogen-based fertilizers is insufficient, and new domestic mineral fertilizer production is required to meet demand. The world also faces a climate crisis, and Sweden is working to reduce its climate impact. Knowledge of Nitrogen-Based Mineral Fertilizer (NBMF) production and the factors affecting its environmental performance is limited in Sweden, which lacks domestic production. The objective of this master's thesis is therefore to provide broad insight for Swedish decision-makers regarding possible variations in urea production, the most common NBMF. To achieve that insight, the study aims to identify key factors shaping urea production's environmental and energy performance across varying Swedish and European conditions, and with it answer whether Sweden offers climate benefits compared to continued European imports.

This study conducts a Life Cycle Assessment of urea production, with the functional unit of 1 tonne of urea ready for market, produced in generalized locations in Sweden and Europe, which are the two options in the factor *Geographical location*. Inquired data reflect industry production and are aggregated into a mass and energy flow audit in a Python program, which also uses assessment data from the Ecoinvent3 database to estimate greenhouse gas emissions and primary energy usage. Potential key factors are identified using an uncertainty identification guide and then filtered for relevance, scope, and impact on the basic Python model (meaning without considering other factors). The chosen subset of factors is then implemented to form the factor-based scenario analysis, structured based on the Design of Experiments.

The study´s findings are presented in a table of potential key factors affecting the environmental performance of urea production. All factors listed are estimated in the factor-based scenario analysis to create 36 distinct scenarios or are provided as rough estimates. *Additional renewable input flows* shift material and energy inputs to the best-performing renewable option from an environmental and energy perspective. It is the most prominent factor that reduces greenhouse gas emissions by an estimated 83% and 91% under Swedish and European conditions, respectively. The same factor also reduces primary energy usage by an estimated 37% and 48% under Swedish and European conditions, respectively. In addition, it has a consistent and significant impact across all scenarios. *Geographical location* also affects greenhouse gas emissions and primary energy, with Swedish conditions emitting 50% ± 10% greenhouse gases than similar European conditions in 2/3 of the scenarios. The geographical difference diminishes and eventually disappears as the share of *Additional renewable input flows* increases. Other factors have a rough estimate of their impact on normal Swedish and European conditions, as executed and discussed. The discussion addresses the knowledge gap regarding the impact of general factors on urea production. The validity of the study´s results is discussed, why the conclusions are trustworthy, and what the uncertainties can and cannot affect.

The conclusion is that 16 factors affecting urea production are identified, however, it is challenging to determine which factors significantly affect the environmental performance of urea. However, *Additional renewable input flows* and *Geographical location* are identified as critical factors, with *Additional renewable input flows* as the most important factor in reducing greenhouse gas emissions and primary energy usage. Construction of domestic urea production is in line with Sweden's goals of fertilizer independence and reduced climate impact.

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List of abbreviations

(g)	Gaseous state
(l)	Liquid state
(s)	Solid state
CH	Switzerland
CO_2	Carbon dioxide
CO_2eq	Carbon dioxide equivalent
DOE	Design of Experiments
GWP	Global warming potential
H_2	Hydrogen gas
IE	Industrial ecology
IS	Industrial symbiosis
kg	kilogram
kWh	kilowatt-hour
LCA	Life Cycle Assessment
LNG	Liquid nitrogen gas
N_2	Nitrogen gas
NBMF	Nitrogen-based mineral fertilizer
NH_2COONH_4	Ammonium carbamate
NH_3	Ammonia
NG	Natural gas
O_2	Oxygen gas
SMR	Steam methane reforming
SMRP	Steam methane reforming pathway
PSA	Pressure swing adsorption
RoW	Rest of the World

1. Introduction

This chapter introduces the subject and the goal of the thesis. It contains the sections Background, Objective, and Research questions.

1.1 Background

The global population is above 8 billion people, an unprecedented number of people in human history (Worldometer, 2025). Projections indicate that the human population will continue to grow and reach 10 billion by 2050, and with it, the global food production needs to double compared to a decade ago (Temegne *et al.*, 2021). It is a challenge for the agricultural sector, which lacks available arable land to expand further, and whose existing agricultural systems already operate at high production levels. Furthermore, the high production levels are associated with major negative environmental impacts and the erosion of agro-biodiversity, further challenging future agricultural efforts (Ibid.).

One prevalent tool to increase food production, utilized for more than two millennia, is the use of fertilizers. They supply crops with essential nutrients the plants need to grow better and give higher crop yields (Marschner and Marschner, 2012). Fertilizers come in different forms, such as bonemeal made from ground bones or nutrient-rich animal feces. However, since the 19th century, industries have supplemented or substituted fertilization with synthesized or mineral fertilizers, which have aided farmers in producing even higher crop yields (Argus Media, 2024).

Sweden uses mineral fertilizers, but currently lacks any domestic production of them (RI.SE, 2023, 2025), and is therefore dependent on imports, primarily from European markets. This dependency is mitigated by employing circular nutrient management practices, such as utilizing manure, biofertilizer from biogas production and wastewater treatment sludge as nutrient substitutes in agricultural systems (Feiz *et al.*, 2022; RI.SE, 2023). While these mitigation efforts help reduce demand, they are insufficient to reduce the total demand for mineral fertilizers. The remaining import needs are something that the Swedish government wants to reduce (Metson *et al.*, 2022; Devault, Woolf and Lehmann, 2024). High volatility in the European mineral fertilizer market is the main driver of this concern, which became apparent early on during Russia's invasion of Ukraine in 2022. Establishing domestic production of mineral fertilizers would lower the necessity for imports, which is feasible (RISE, 2023; Rodriguez, 2023).

There are several distinct types of mineral fertilizers that can be produced, commonly categorized according to their primary nutrient content. In Sweden, as well as globally, Nitrogen-based mineral fertilizers (NBMF) are sold in the largest volumes, along with phosphorus-, sulfur, and potassium-based fertilizers (SCB, Statistiska centralbyrån, 2025). 185 thousand tons of NBMF were imported by Sweden in the 2023/24 period (RISE, 2023; SCB, Statistiska centralbyrån, 2025). Sweden can in theory, meet most of the requirements for phosphorus and potassium within its borders through circular nutrient management (Metson *et al.*, 2022). This is not the case for nitrogen-based fertilizers, which have significant atmospheric and aquatic losses, leaving maximum nutrient recycling at 44%. This makes NBMF production a relevant focus for further investigation.

The element nitrogen is plentiful, more so than any other element in Earth's atmosphere. However, it primarily exists as nitrogen gas (N_2), a molecule that most plants are unable to absorb. Instead, plants absorb nitrogen mainly in the form of ammonium or nitrate compounds. Ammonia, synthesized from atmospheric N_2, serves as the starting point of all NBMF. While similar conversion of N2 to plant-accessible nitrogen occurs naturally, approximately 29% of global conversion of N2 now occurs due to humans (IEA, 2021). Urea stands for 55% of all industrially produced ammonia in 2019, and is by far the most commonly used mineral fertilizer, with 90% of all NBMF being urea (Bargiacchi, Antonelli and Desideri, 2019; IEA, 2021). Worth noting is that industrial ammonia is primarily synthesized by H_2 and N_2 (Bargiacchi, Antonelli and Desideri, 2019).

There are existing works related to newer and more sustainable methods for producing ammonia and urea, which can aid Sweden´s goal for a lower climate impact (Rodriguez, 2023). These newer production methods are typically indicated or evaluated to be sustainable using Life Cycle Assessments and are essential for comparing environmental performance between products or systems (IEA, 2021; Feiz *et al.*, 2022; Ammenberg, 2023; Rodriguez, 2023). One relevant example of system evaluation occurred in Rodriguez (2023), in which urea production pathways were studied, as illustrated in Table 1.

Table 1.Identified industrial production technologies in the production system of urea, based on technological pathways for urea mineral fertilizer production by Rodriguez (2023)

Nitrogen	Cryogenic separetion	Pressure swing adsorption	
Hydrogen	Methene steam reforming	Water electrolysis	High-thermal plasma
Ammonia	Haber-Borsch	Non-thermal plasma	
Urea	Borsch-Meiser		

Rodriguez (2023) looked at sets of production methods and compared them. Getting good, relevant information, such as the H_2 production process uses more primary energy but can be more sustainable, depending on the electricity source. Such insight is paramount if Sweden is to meet its greenhouse gas reduction goal. Nevertheless, it is still insufficient for decision-makers to gain an insightful and broad understanding when instructing the development of a urea production site. There is limited understanding of how different factors, such as the electricity mix, interact with other prominent factors, such as different production technologies or additional material recycling methods, to shape greenhouse gas emissions and primary energy performance. However, calculating all factor interactions and presenting them is neither feasible for a thesis nor would the results be comprehensible, as bloated results would dilute the core of the matter. Instead, prioritizing the use of uncertainty analysis and conducting uncertainty propagation allows for a more accurate and robust assessment (Batstone, 2013; EFSA Scientific Committee *et al.*, 2018).

There is a demand for information policies to facilitate the design and construction of effective urea production sites that can meet the growing need for domestic NBMF production and Sweden's ambitions to reduce greenhouse gas emissions and dependence on fertilizer imports.

Such policy guidance can be developed through a systematic evaluation of the urea production system under varying technical and contextual conditions that affect environmental performance. Understanding how key factors interact and influence energy and environmental performance is central to this effort.

To support this need, this thesis develops a modelling framework capable of producing values for different feasible real-world conditions for analysis. The results are intended to provide insights into possible production in Sweden and potential advantages compared to European imports. Furthermore, it should identify key technologies and conditions when planning an industrial urea production system. Enabling readers and decision-makers to gain a broad system-level understanding of possible pathways forward. Rather than identifying production routes and comparing them, this study focuses on understanding the impact of factors and their interactions on a production's environmental impact.

1.2 Objective

To evaluate the global warming potential and energy usage associated with urea production under varying conditions in Swedish and European contexts, with the purpose of assessing environmental justification for domestic production in Sweden. The objective is further to highlight the influence of key factors on environmental and energy performance, and to provide decision-relevant insights into how domestic urea production compares to European. The aim is also to assess which factors should be prioritized to improve greenhouse gas and energy performance.

1.3 Research questions

1. What are the key factors that impact greenhouse gas emissions and primary energy usage for urea production?

2. How do the geographical locations of Sweden and Europe differ in their greenhouse gas emissions and primary energy usage for the production of one tonne of urea, and how does factor interactions affect the performance?

2. Theoretical framework

This chapter introduces theoretical concepts and methodological approaches applied in this study. The concepts of resource efficiency and industrial symbiosis between industries are presented, leading to Life Cycle Assessment (LCA), and its function as a sustainability evaluator for environmental and energy performance. The system perspective explains how LCA is applied to an entire system, integrating lessons from LCA and Industrial Ecology. In addition, the chapter addresses uncertainty analysis as a means of identifying and screening potential factors and introduces uncertainty propagation to explain how computable models can be structured and applied to conduct a successful factor-based analysis.

2.1 Resource efficiency and industrial symbiosis

Resource efficiency refers to the optimal utilization of natural resources, maximizing the benefits derived, whilst minimizing waste generation and subsequent environmental impact (Wulf, Ansari and Nyhuis, 2013). One concept to increase efficiency is called Industrial ecology (IE), which holds that efficiency and reduced environmental impact are achieved by mimicking ecological systems in industrial production through recycling mass and energy. The name Industrial symbiosis (IS) arrived later, and is the practical manifestation of the concept IE (Lifset and Graedel, 2002). By mapping mass and energy flows for each production, discarded or unused materials and energy can be recycled or reused as input materials for other productions. A well-documented example of IS is Kalundborg Industrial Park in Denmark and Sotenäs in Sweden, which, despite different resources, still have successful IS networks (Lifset and Graedel, 2002; Martin, 2018). The principle of resource recycling and efficiency is relevant to model development and to identifying factors that affect environmental and energy performance.

2.2 LCA and system perspective

The LCA method is designed to evaluate the impact of industrial production and service systems. The assessment is based on the compilation and analysis of inputs, outputs, and the potential environmental effects throughout a product's life cycle. A typical life cycle is cradle to grave, however, a cradle to gate is common for products with comparatively little impact during use and end of life stage to their production stage. The cradle to gate perspective can be an option for certain actors seeking to reduce total impact and emissions, however, this is not the case for this study. This means that impacts from materials and energy used in production are assessed, but nothing is considered after the product reaches the gate. The life cycle is centered on a functional unit in order to evaluate production performance (ISO, 2011; Ammenberg, 2023). A final product is typically a good functional unit, especially when mass flows vary across processes. Which fits the description of urea production (Ammenberg, 2023).

An LCA comprises four phases. The first phase, called "Goal and scope definition", defines the goals and scope. The functional unit is also defined. The second phase, "Life cycle inventory", entails the systematic collection and processing of data on process input and output flows. The third phase, "The impact assessment", involves classifying and characterizing the collected data by environmental problem categories and the representative emission measurement units associated with each category. For example, X kilograms (kg) of methane is burned to produce one functional unit, which corresponds to Y kg of carbon dioxide equivalents (CO_2 eq) in the

category global warming. The last phase is evaluation and reporting. The evaluation and reporting phase is key for this study, since evaluating uncertainties is part of its purpose. Reporting requirements are also paramount, as academic work such as this study requires them (Ammenberg, 2023).

LCA provides a comprehensive perspective for assessing environmental performance in a poly-industrial process, and with a system perspective provided by IE, reveals otherwise unseen opportunities (Ayres and Ayres, 2002; Ammenberg, 2023). An example could be a CO_2 purge outlet containing high-purity CO_2 being moved from company A to company B as input to their greenhouse to improve plant growth. LCA also involves challenges that should be considered when conducting the assessment (Ayres and Ayres, 2002; Ammenberg, 2023). One key challenge is what to include and, more importantly, what to exclude from a system boundary. A common mistake is to create a scope that is too wide, with too many elements included, making the assessment too taxing (Ammenberg, 2023). Then there are the more universal assessment challenges that deal with data availability and quality (Reap *et al.*, 2008; Schmidt, Götze and Sygulla, 2015; Ammenberg, 2023). The broader systems thinking inspired by IE guides decisions on boundary width and whether to broaden or narrow the system.

According to ISO 14044, allocation of energy and climate impact outside the system boundary should, if possible, be avoided. This can be done by extending the boundary parameter and introducing recovery technologies. If the allocation is not feasible within the boundary system, a fair assessment is necessary. If there is a physical connection between input and output flow, then the allocation should be divided into products or functions, based on the underlying physical relationships, such as energy or mass (Ekvall *et al.*, 2025).

2.3 Uncertainty analysis

Uncertainty analysis is an essential tool for environmental assessment, facilitating decision-makers´ informed and reliable choices (Batstone, 2013). The analysis achieves this by extensively identifying and quantifying uncertainties throughout the study object. To help the assessor identify uncertainties, key areas are brought up: data, models, and expert judgements. *Data-related* uncertainty arises from empirical evidence obtained from experiments or measurements. The uncertainty area, *model*, is the umbrella term for uncertainties related to mathematics, statistics, or logical relationships between inputs and outputs. For example, the functional unit often normalizes impacts to a per-unit measure, despite production performance varying with batch size or production interval. The last uncertainty area, *expert judgement*, contains an impactful assessment provided by qualified individuals. There is a hierarchy where *data* is considered the most secure *area* and *expert judgment* the least secure (EFSA Scientific Committee *et al.*, 2018).

The identification process classifies uncertainties into standard and non-standard uncertainties. Standard uncertainties are uncertainties that clearly are impactful for the assessor or are explicitly stated to be impactful in previous works. This insight is usually acquired when gathering information from other studies or databases. Non-standard uncertainties are deviations from standard procedures and studies. It can be statements or data that lack qualitative references or reasoning. These kinds of uncertainties are often case-by-case (EFSA Scientific Committee *et al.*, 2018).

There are a handful of approaches to uncertainty analysis, and the correct one depends on the assessment case. One procedure type is case-specific assessment, which is recommended when no prior assessment has been conducted. It instructs the assessor to systematically examine every *area* for uncertainties. The last classification step is recommended to be performed without always correctly aligning all uncertainties, since this can be inefficient.

2.3.1 Uncertainty propagation

Uncertainty propagation is a critical aspect of uncertainty analysis. It is typically applied to computational models, such as Design of Experiments (DoE), to assess how input variability affects output in complex systems (Batstone, 2013; Liu, Jiang and Xiao, 2020). It evaluates the impact of uncertainties in data or assumptions that affect the model's outputs. These factors often take the form of numerical multipliers or parameters introduced to account for variability. For instance, differences in CO_2 emissions from the same type of fuel, or variations in compressor efficiency. Safety factors are applied to compensate for a greater variability in input data, such as when two datasets largely differ or when the source of the factor is unreliable (Liu, Jiang and Xiao, 2020).

There are several methods for uncertainty propagation, including Polynomial Chaos Expansion, Moment-Based Approach, and Monte Carlo Simulation. The last-mentioned Monte Carlo Simulation is typically used for its specialization in robustness and flexibility, which is well-suited to this study. The method originally involves repeated sampling of values from defined parameter ranges of distribution and probabilities. The iterations generate a spectrum of probabilistic outputs that can provide insights into variability (Liu, Jiang and Xiao, 2020; Esguerra *et al.*, 2021).

2.4 Modeling structure for variation-rich assessments.

DoE is a system-building structure suitable for uncertainty propagation since it uses factors to generate factor-based scenarios. DoE also screens factors and deduces the critical ones. The design method typically consists of three major steps. Step A: detail the model structure, which is the creation of **n** datasets with **m** factors influencing all of them, as shown in Table 2. The number of scenarios is then up to $\mathbf{m^n}$ (Esguerra *et al.*, 2021).

*Table 2. Creation method of scenarios based on Datasets **n** and Factors **m**, based on Step A in the DoE process, by Esguerra et al. (2021).*

Factors (m)	Datasets (n)				
Factor 1	1, 1	1, 2	1, ..	1, ..	1, n
Factor 2	2, 1	2, 2	2, ..	2, ..	2, n
Factor ..	.., 1	.., 2	.., ..	.., ..	.., n
Factor ..	.., 1	.., 2	.., ..	.., ..	.., n
Factor m	m, 1	m, 2	m, ..	m, ..	m, n

Step B conducts scenario assessments using LCA or another evaluation method, such as Life cycle costing, or a combination of assessment methods, to analyze the scenarios created in Step A based on their energy and mass flows. The last step, Step C, performs a sensitivity analysis to systematically determine the critical performance factors. Analytical methods include Process hotspot analysis, Trade-off analysis, Eco-efficiency analysis, and Global sensitivity analysis (Sauve *et al.*, 2023)

3. Method

The method chapter outlines the systematic approach applied in this study to ensure transparency and reproducibility. It describes the data collection procedure guided by the selected research philosophy. As well as the methodological steps in implementing a Life Cycle Assessment (LCA) on mass and energy flows related to production technologies necessary for urea production, with a cradle-to-gate life cycle perspective. Finally, the steps leading up to the factor-based scenario analysis are described, along with how it examines Swedish and European conditions under varying factors.

3.1 Data collection

In order to extract data reflecting the real world whilst still producing results in a timely fashion, the research philosophy of Postpositivism was adopted as a tool. Postpositivism is a combination of pragmatism and constructivism; it prioritizes solving problems, whilst adhering to qualitative data and the view that reality can be approximated through knowledge, data, and objectivity (Peggy Shannon-Baker, 2023). This study adhered to postpositivism and used it as a guide for data collection and for implementing the hierarchy of evidence. From peer-reviewed literature to gray data sources to independent web pages. The philosophy is also pragmatic, meaning the research could maintain momentum even when data in one hierarchy were lacking. A popular science page dedicated to physics, such as engeneringtoolbox.com, was used primarily for data that articles seldom use and companies have behind paywalls. It was prioritized as a search engine after proving so practical. ScienceDirect produced hundreds of thousands of search hits at the time, which were not applicable, and only the first 20 or 40 titles and abstracts were browsed before moving to another search engine. Table 3 illustrates the general process of finding data for the study and the model.

Table 3. "Database" and "keywords" that were used to find articles are shown in the first two columns. The "Database hits" show the number of articles found by the search. Searches on ScienceDirect returned hundreds of thousands of hits, with the most relevant appearing first. Therefore, only the first 20 or 40 article hits were included as an option for the next step, "Selected from title & abstract", which selected articles to read thoroughly based on title or information in the abstract. The last column, "Selected," shows the number of articles used in this study. Others refers to smaller databases with either similar or gray literature. The databases are Jordbruksverket, RI.SE, and the data from Linköping University: courses and professors*

Database	Keywords	Database hits	Selected from title & abstract	Selected
Science-Direct		1100 000	16	4
	Ammonia production			
	Nitrogen mineral fertilizer			
	Urea production			
	Climate impact			
	Resource efficiency			
	Uncertainty propagation			
	SMRP + Hydrogen production			
Scopus		245	87	21
	Green production			
	Urea production			

	Resource efficiency			
	Sustainability			
	Industrial ecology			
	Industrial symbiosis			
	Gas vs solid transportation			
	Why LNG and not NG pipelines			
	Why is uncertainty analysis important			
	Water electrolysis hydrogen production			
	Bargiacchi, Eleonora			
	LCA of H_2 production in industry			
	Range of temperatures for low, medium, and high industry waste heat			
	Production efficiency dependent on volume in urea production			
Others*		303	47	16
	Mineralgödsel			
	Kvävegödsel			
	Nationell produktion			
	Förberedskap			
	Priser			
			Total	**41**

3.2 Use of AI

This thesis used advanced AI tools, namely: Scopus AI, Copilot, Grammarly, and ChatGPT. For Scopus AI, it was used to find articles and relevant information. All information from the AI prompt was verified in the abstract article or in the article. Copilot and ChatGPT met the same needs: guiding the text to be more formal and suggesting phrasing and structural changes. Copilot was also used to help find where data was located in articles by uploading the article's PDF and asking the AI where specific information was. It is aided by limiting the time spent rereading articles to verify or reread information known to be there. For example, the author knows that a table shows great values of hydrogen production flows, so the location is requested, and Table 2 is the response. The information consistently originates from the article, and its use is akin to a more localized search engine or to the search function in documents, with word-level comprehension. Grammarly, Copilot, and ChatGPT were used for text refinement, with manual implementation of word suggestions and bolded grammatical mistakes.

3.3 Goal and scope definition

The goal is to achieve the study´s objective by answering the research questions, detailed in sections 1.1 and 1.2. They can be summarized as identifying key factors, comparing results, and examining key factor interactions in the production of urea in Sweden compared to Europe. The comparison between the two geographical locations focuses on their environmental impact and energy demand, specifically global warming potential and primary energy usage. One factor integrated into the model at the beginning was production pathways, which will be defined promptly, and that is why 1,000 kilograms (kg) or 1 tonne of produced urea ready for market is a suitable functional unit.

The scope has a cradle-to-gate life-cycle perspective, since the finished product is assumed to affect the *Use* and *End* phases equally. The production process is the objective of comparison and includes activities within the foreground system, as illustrated in *Figure 1*. Note that the focus is on active production flows that affect greenhouse gas emissions and energy use. Supporting activities, such as electricity for lighting or paper for contracts, are excluded. Processes that do not significantly affect greenhouse gas emissions or energy usage are therefore also outside the foreground system, such as supporting activities like lighting, component storage, and material movements, between production technologies. The input material and energy flows to the active production, and the outgoing pollutants are assessed using flow data, assumed negligible due to no or minimal impact, or assumed. CO_2 is the only assumed flow, with it being equal to CO_2 eq. By-products are not included in the basic impact assessment, small system expansions will occur as factors are introduced.

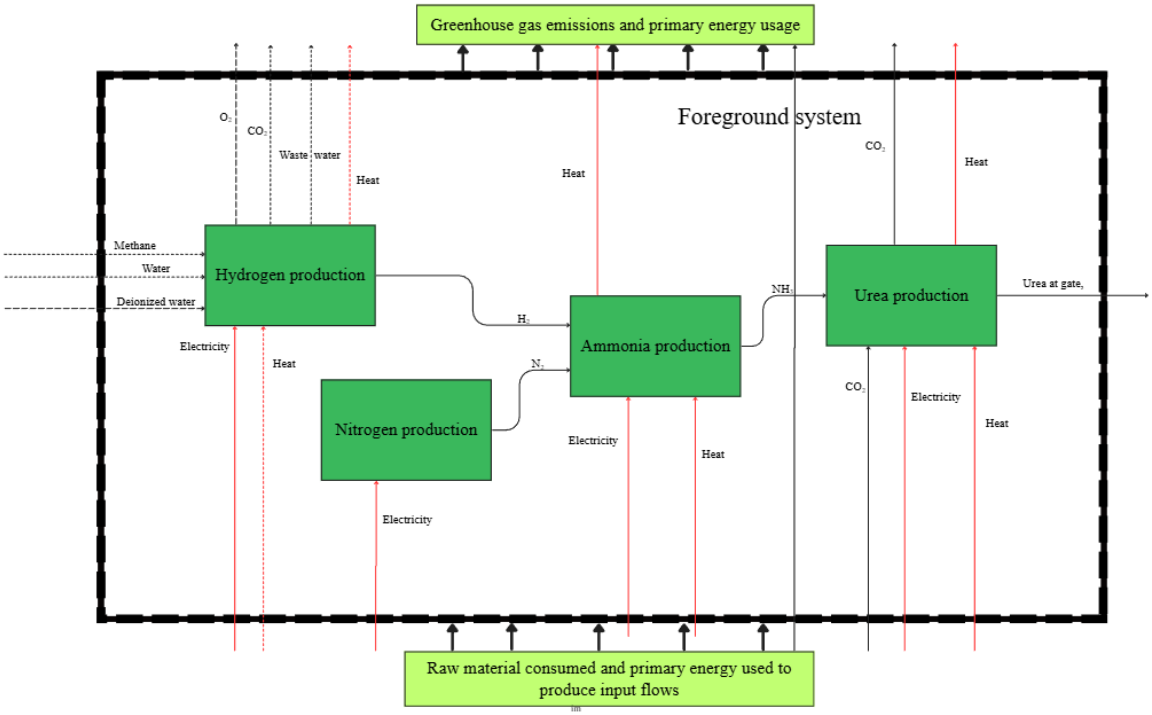


Figure 1. Shows the calculated flows in the urea production process. Input flows origin, their impact during urea production is calculated together with the outputs from the production process, excluding the finished product. Nature green boxes indicate the production technologies that produce the main input flows tied to the functional unit. Red arrows are calculated energies and black arrows are calculated mass flows, meaning the model will take them into consideration when evaluating. Note that H_2 production has two different technologies with different inputs and outputs. The full line is the type-flows they have in common, the dashed arrows are unique type-flows to the electrolysis production method, and the dotted arrows are exclusive to the team methane reforming production method. The system boundary ends for the urea at the Swedish gate, ready for market, since the differences in impact are negligible after that point.

The general assumption for the production of urea in all production technologies is that the room temperature is at 25 ºC, and that compressor efficiency is at 85% (Chisalita, Petrescu and Cormos, 2020; García *et al.*, 2025).

All production is assumed to be at one site, with water, industrial-grade carbon dioxide (CO2), and natural gas (NG) entering the system as mass. The reason for the one-site production system

is that urea production is usually connected to ammonia plants and their closeness in production purpose (Bargiacchi, Antonelli and Desideri, 2019; IEA, 2021). Hydrogen gas (H_2) and nitrogen gas (N_2) are assumed to be produced on site based on three arguments. The first is that gas transportation requires pipeline infrastructure, which is inflexible and ill-suited for long-distance transportation (Ristic, 2008; Ulvestad and Overland, 2012). The second is that liquid transportation of gases requires significant cooling energy to liquify, such is the case for H_2, and to a lesser extent, N_2 (Incer-Valverde *et al.*, 2023; Engineering ToolBox, 2025d). The last reason is the wide availability of industrial H_2 and N_2 production, including for small-scale operations (O'Neill industrial, 2020; Vätgas Sverige, 2026).

After studying the maps of both NG networks and biogas facilities, the study chose to forego such calculations. This is because the map tells that Sweden and Europe, as a whole, have similar transport distances and that the NG infrastructure is complex (Lustenberger *et al.*, 2019; Gas infrastructure Europe, 2025). It is slightly different inside Sweden, with the gas grid only reaching as far as Gothenburg, so gas tanks or liquid natural gas (LNG) would have to cover the rest of the travel, causing more emissions and primary energy usage. On the other hand, there are several port entries for NG on the west side of Europe, likely receiving boat shipped LNG from the United States of America or another distant nation, and emitting considerably more greenhouse gas emissions than average due to the transport. (Council of the European Union, 2024). Sweden gets its natural gas in pipelines from Denmark, which is hooked up to the European natural gas network pipeline (Energimyndigheten, 2022). Since emissions for both Sweden and Europe overall are considered, it is assumed that they both source their natural gas from the same place. Greenhouse gas emissions and primary energy usage are calculated as the average per kg methane from the four top suppliers of NG to Europe, whose supply each reaches above 10% of the total NG and LNG import (Council of the European Union, 2024). That means the top supplier contributes more to the average impact than the second top supplier, from a volume perspective.

Up to this point, the scope has encompassed *basic* Swedish and European conditions, as well as production pathways. However, no robust conclusions can be drawn from a single set of assumptions. Robust insights can be gained by creating scenarios from factors, in other words, as a set of assumptions with accurate industry values.

3.3.1 Factor and scenario definitions

The study used the guide from the European Food and Safety Agency´s Scientific Committee to identify and categorize factors into standard and non-standard uncertainties using a case-specific method. Uncertainty identification began with the flow-data collection of the *Basic conditions*, and explicitly stated uncertainties were collected from academic articles. When concluded, a holistic uncertainty identification was performed for each *area*.

The identified uncertainties were named potential factors. However, non-standard uncertainties were omitted from deeper estimation, unlike the standard uncertainties, on the basis that their expected impact did not exceed this threshold and because their inclusion would have required additional assumptions that would not substantially enhance the analytical outcomes. Their exclusion is recognized as a methodological limitation and further discussed in Section 5.2.

Each standard uncertainty had its potential impact estimated and filtered to identify key factors that were integrated into the scenario analysis. Factors that provided overlapping information or whose effects were already captured by other included variables were removed to maintain a clear and efficient model structure. For example, the *Electricity mix* was excluded because its influence is already captured by *Geographical location* and variation in *Additional renewable input flows*. The distance between production plants was also excluded, as this factor falls outside the defined system boundary, as detailed in Section 3.3 and addressed in Section 5.2. Factors were additionally excluded based on impact. The factors were given a reasonable range based on the findings, then directly and temporarily integrated into the code to form the *Basic conditions.* If the influence was less than 20%, they were excluded. An example is *H_2 capture efficiency in electrolysis*, which identified a H_2 capture rate of 72% to 100%, and integrated to assess its impact manually by changing the H_2 gas production to match possible losses or gains in production, so that it would still produce the same amount of H_2 required for the rest of the urea production.

The factors used in the factor-based scenario analysis are presented in Table 4

Table 4 Factors that are used in the factor-based scenario analysis. The first column lists the factors' names. The second column presents the number of parameter options and their name. The third column concisely describes the factor. Note that production pathways share the production technologies, except for H_2 production.

Factors	**Parameter options**	**Short description**
Geographical location	Sweden, Europe	Changes the input sources to those of the geographical location
Production pathways	*SMRP*, *Electrolysis*	Different urea production technologies used in the production pathways. SMRP uses the H_2 production technology of Methane steam reforming, and Electrolysis uses water electrolysis H_2 production technology
On-site Industrial Symbiosis (IS)	*No on-site IS, 50% on-site IS, 100% on-site IS*	Increases the recycling of materials and energies within the foreground system.
Additional renewable input flows	*No extra renewability, Partial renewability increase, Full Renewability*	Changes the input sources from established options to the most renewable and environmentally friendly sources.

Factors were integrated into the model in a way inspired by Design of Experiments (DOE), as seen in *Figure 2*

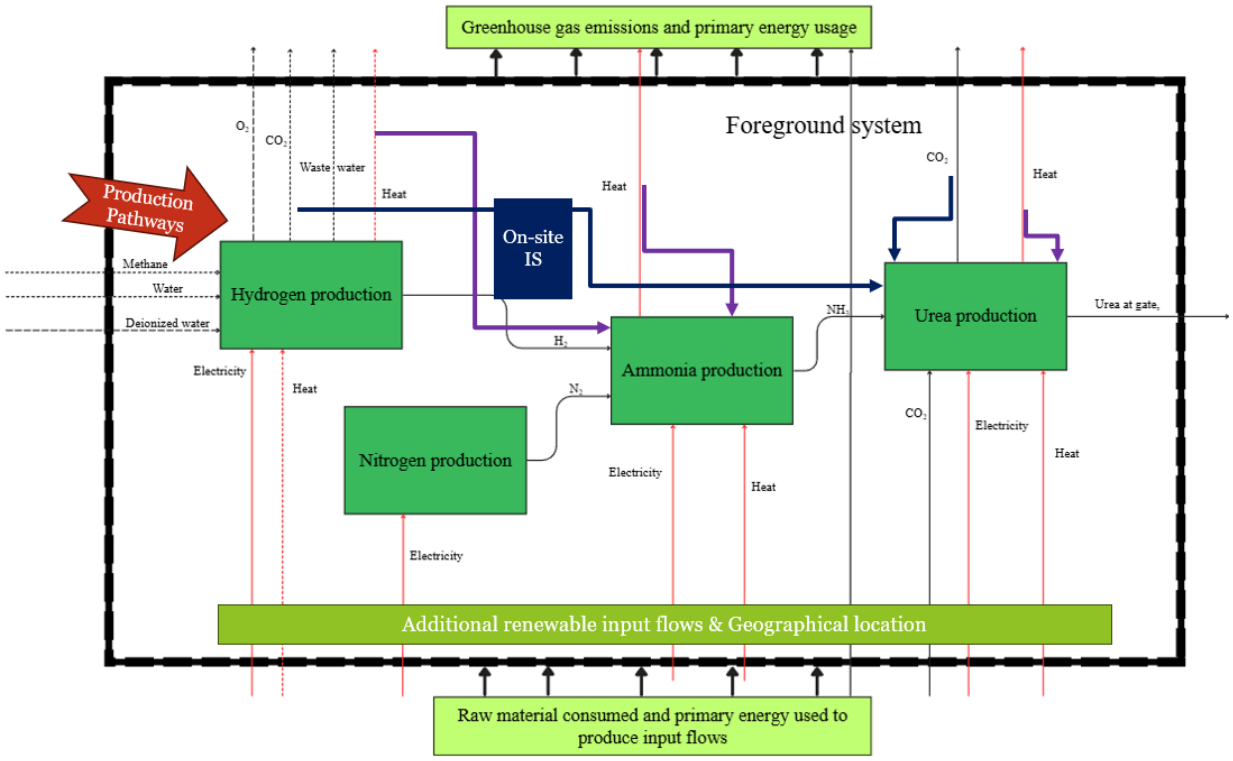


Figure 2. Shows the calculated flows in the urea production process, along with how factors impact. Input flows origin is affected by both Additional renewable input flows and Geographical location, and their impact during urea production is calculated together with the outputs from the production process. Nature green boxes indicate the production technologies required to produce the functional unit. The input flows are indicated by red arrows for energy and black for materials flows. Note production pathways change the H_2 production, and they have different inputs and outputs. The full line is the type-flows they have in common, the dashed arrows are unique type-flows to the electrolysis production method, and the dotted arrows are exclusive to the team methane reforming production method. The system boundary ends for the urea at the gate, ready for market, since the differences in impact are negligible after that point.

. Instead of using probability distributions or randomization, each factor was assigned two or three parameter values. The binary values were alternatives, whilst the three parameter values had the structure of a minimal, a maximal, and a representative midpoint value. Together, the number of scenarios calculated from the factor parameters is given by Equation [1]. This totals 36 distinct scenarios.

Equation for the number of scenarios analyzed.

$$Scenarios = \mathbf{2}^{f} * \mathbf{3}^{f} \quad \textbf{[1]}$$

Geographical location uses different sources of material and energy flows, which impact environmental metrics differently. Generalized location data is used, for example, the national electricity mix of Sweden is used for the Swedish context, and the European electricity mix is used in the European context. How the factor affects the assessment of flows is detailed in Appendix 1.

Production pathways and their assumptions are specified in Section 3.4, where all production technologies are presented. *SMRP* indicates *Basic conditions,* and the difference between the

production pathways is that *SMRP* uses the H_2 production technology of Steam methane reforming, and *Electrolysis* uses Water electrolysis instead.

On-site IS recycles mass and energy flows internally between production technologies and within the foreground system, shown in *Figure 1*. The parameter has three options, as shown in Table 4. *No on-site IS* indicates *Basic conditions* that no IS occurs between production technologies, *50% on-site IS* has and *100% on-site IS* recycles the corresponding percentage of possible outgoing material and energy back into inputs. The output flows are reduced by the same amount as the inflows and cannot be recycled if there is no input demand flow. The recycling is assumed to function without additional energy or mass usage.

Additional renewable input flows are the other factor, apart from *geographical location*, that affects how flows are assessed and is similarly presented in Appendix 1. The first parameter option, *No extra renewability*, is used under *Basic conditions* and assesses the flows with normal generalized sources of mass and energy that enter the foreground system in *Figure 1*. At *Full renewability*, flows are substituted for the most renewable option in the assessment database Ecoinvent3 for that *geographical location*. However, the assessment data from Ecoinvent3 is not always that exact, it does happen that some values are used in all conditions. This can be observed in Appendix 1. *Partial renewability increase* is the middling value and uses the average between *No extra renewability* and *Full renewability*. It is assumed that the renewable flows can be found within a reasonable distance, for example, biogas has an assumed transportation distance of 1000 km.

3.4 Life Cycle Inventory

The goal of evaluating urea production requires data, which was gathered using the Life Cycle Inventory method to collect the material and energy flow data necessary. The input and output flows of H_2, N_2, ammonia, and urea for each production technology will be presented here. The flows collected are then combined into the two *production pathways*. The texts below are primarily the work of the articles, although the data often required adjustments to be inside the foreground system or for the sake of accuracy.

3.4.1 Hydrogen gas, Steam methane reforming

The classic steam methane reforming (*SMR*) is one of the most established industrial methods for hydrogen production (Lee *et al.*, 2021). An example of an industrial SMR system for H_2 production is shown in *Figure 3* (Lee *et al.*, 2021), however the article lacks the required data. This is why data from Chisalita *et al.* (2020) have been used to calculate mass flows and electricity demand for the model. The essence of the industrial SMR production system is illustrated in *Figure 4*.

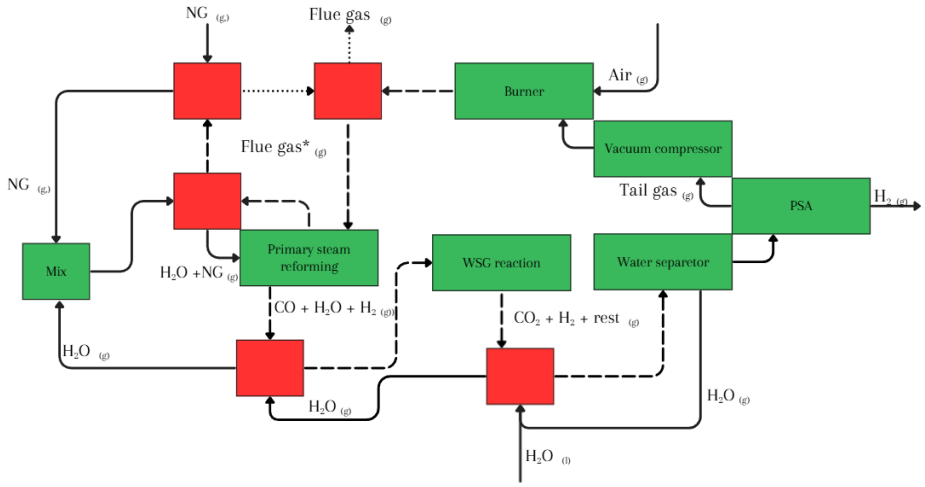


Figure 3. Shows an example of a full steam methane reforming production process of H_2, based on Lee et al. *(2021). All arrows are mass flows. Red boxes are illustrations of heat exchangers, i.e., no mixing of mass flows. Heat is generated by the burning of tail gas and air, as seen in the top-right corner. The arrow network is extensive, but the rule is that the same dashed, dotted, or bolded line that goes in, goes out in the arrows with the same markings. The flue gas indicated by * goes clockwise around the label before exiting.*

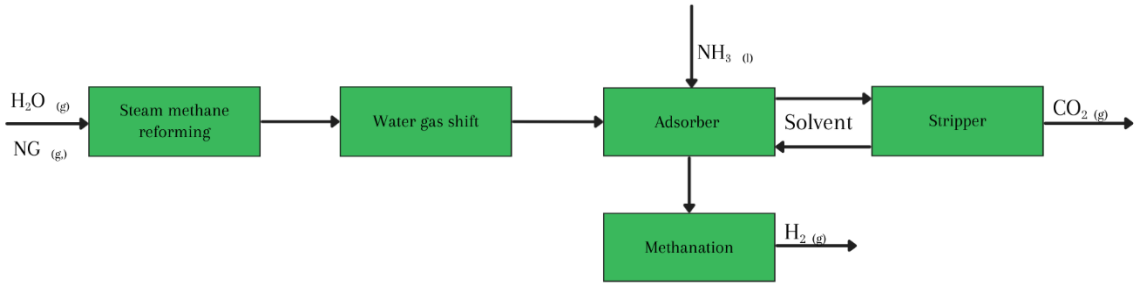


Figure 4. Shows an illustration based on Chisalita et al. (2020) description. Note that it does not show all the processes involved in the production of H_2, per the study's scope. The black arrows are mass flows. The ammonia mass flow is small and used as a solvent.

As can be seen in *Figures 3* and *4*, the systems differ at the end with Lee *et al.* (2021) using PSA and Chisalita et al. (2020) using methanation as a recovery system. Chisalita et al. (2020) has water vapor mix with natural gas at high temperature and pressure to decompose methane, as seen in Chemical equation [4], then react again in the Water gas shift, as seen in Chemical equation [5]. The flows and numbers that Chisalita et al (2020) comes with a caveat. The inflows of mass to the SMR process are assumed to already be at 29 bar and around 900 ºC.

Using the known mass flows, this study calculates the heating and compression energy required, along with the assumptions detailed in Appendix 3. Ammonia is not considered by the model, since it has a relatively insignificant mass flow of less than 1% of the produced H_2, and requires little energy for cooling, being at 5 ºC (Chisalita, Petrescu and Cormos, 2020). The input values to the model alter depending on factors, however, the input and output flows that the model calculates are in Appendix 2. Chisalita et al. (2020) has 1 heat flow leaving the system at 40 ºC, however, that is a low value heat and is assumed to be negligible for this study.

Chemical equations [4] and [5], all molecules are in gaseous form.

Methane and water vapor to carbon monoxide and H_2

$CH4 + H_2O \rightleftharpoons CO + 3\ H_2$ [4]

Carbon monoxide and water vapor to CO_2 and H_2

$CO + H_2O \rightleftharpoons CO_2 + H_2$ [5]

3.4.2 Hydrogen gas, Water electrolysis

Water electrolysis is a simple industrial method to produce H_2, however, it is energy-intensive. The steps are shown in *Figure 5*. Deionized water enters at room temperature, where it is blasted with electricity and then subjected to high pressure. Which breaks down the water, as seen in the Chemical equation [6]. H_2 leaves in the gaseous form and at room temperature, when the product is ready (Valente *et al.*, 2020). There is potassium hydroxide with a relatively low mass flow of 0.85 g needed to produce 1 kg of H_2, so the impact is assumed to be negligible. Valente *et al.* (2020) has its flows in the same units as the model, which makes integration possible without conversion, the flows used can be seen in Appendix 2 (Valente *et al.*, 2020).

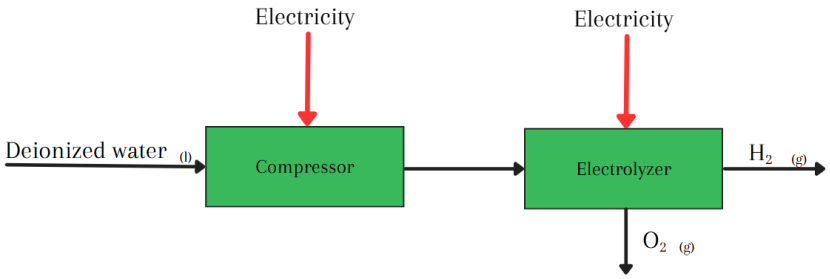


Figure 5. H_2 production via water electrolysis, based on Valente et al. (2020). Red arrows show the flow of energy, and black arrows show the flow of mass. The unnamed arrow is pressurized deionized water.

Chemical equation [6]

$2\ H_2O_{(l)} \rightleftharpoons 2H_{2\,(g)} + O_{2\,(g)}$ [6]

3.4.3 Nitrogen gas, Pressure swing adsorption

Pressure Swing Adsorption (PSA) is an established method for N_2 production. The industrial PSA system relies on selective adsorption of air components, primarily oxygen and water vapor, onto adsorbent materials, allowing N_2 to pass through (Lewis, Robert, 2012). This process requires a higher pressure of between 6 and 10 bar, 8 bar was used in this study. The system is shown in *Figure 6*.

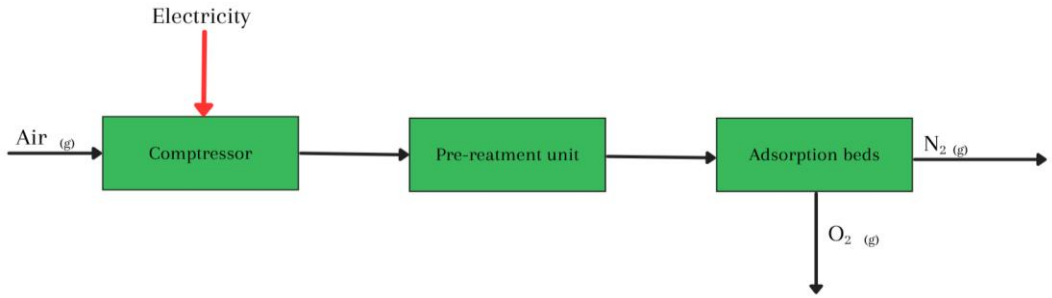


Figure 6. N_2 production via Pressure swing adsorption, based on Lewis, Robert (2012). The red arrow shows the input of electricity, and the black arrows show the flow of mass. The unnamed arrows are pressurized air.

The resulting purity of N_2 depends on the adsorption system's efficiency, with typical values ranging from 95% to 99.9995% (Lewis, Robert, 2012). The energy requirement varies with the purity needed (Syakdani *et al.*, 2019). The industrial N_2 production method is simple and well-established, so few current articles provide the mass and energy flows required by the model. The vast majority of the required electricity for the PSA method is from the compressor, and has therefore been allocated the full energy usage (Jüntgen, Knoblauch and Harder, 1981; Syakdani *et al.*, 2019). Product specification data was therefore obtained from a company website, where current technological flow data could be found for varying purity requirements (O'Neill industrial, 2020). Purity and energy consumption go hand in hand, so the best case is to achieve the required purity and no more. Impurities such as oxygen and water vapor can damage the next Haber-Bosch process and should therefore be below the 10 parts per million limit, requiring a purity of above the 98%. The energy demand for 99.5% was therefore used in this model. The energy requirement to fill the system with air is assumed to be zero. The study likewise assumes that oxygen gas (O_2) does not affect climate impact in any measurable way when it is released back into the air. The two flows that have been mentioned to be calculated are detailed in Appendix 2.

3.4.4 Ammonia, Haber-Bosch

The Haber-Bosch process is another highly established process, accounting for about 90% of all ammonia production (Bargiacchi, Antonelli and Desideri, 2019). The main flow is a pressurized mixture of H_2 and N_2, via a flash separation, to create ammonia (NH_3). The chemical equation [3] illustrates the molecular change, and an industry example is illustrated in *Figure 7*. The model considers efficient processes by calculating in and outflows of mass and energy. Bargiacchi, Antonelli and Desideri (2019) produce a Sankey diagram showing some of the in- and outflows of energy stemming from the input value of 1kilo-mol per second of H_2 as the basis. It is not a single-pass process as for urea, however, the low recovery rate of H_2 and N_2 could have been diminished with a recovery and separation technology, such as Pressure swing adsorption (PSA). Nevertheless, a recovery technology is not calculated, which is a limitation of this study (Erfani, Baharudin and Watson, 2024). The functional unit from Bargiacchi, Antonelli and Desideri (2019) was converted to per kg ammonia, and integrated as ammonia input for the Bosch-Meiser urea production. This did not include purge recovery technologies. The model stores the flows as either kWh per kg ammonia or kg per kg ammonia. The Inputs and Outputs can vary depending on factors and the flows that were calculated can be seen in Appendix 2 (Bargiacchi, Antonelli and Desideri, 2019).

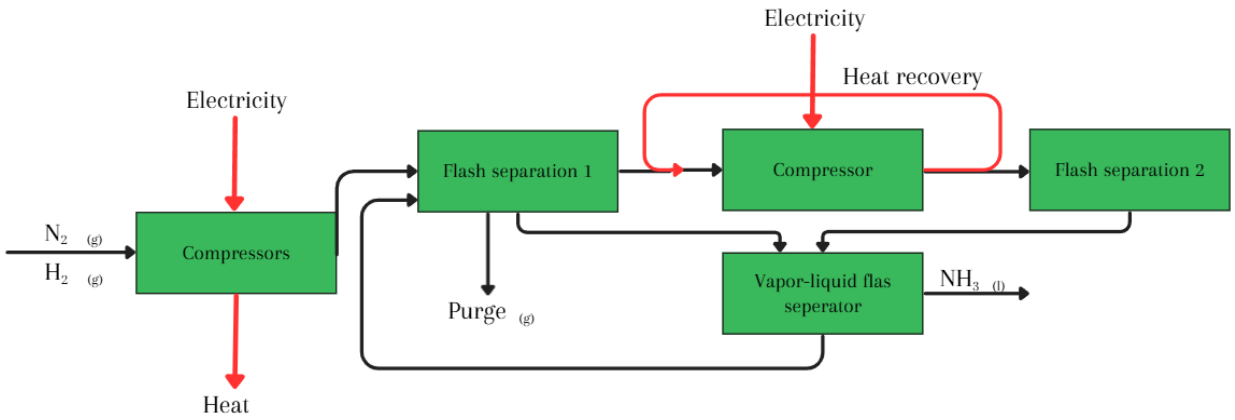


Figure 7. Internal flow of the Haber-Bosch process, based on Bargiacchi, Antonelli and Desideri (2019). Red arrows show the flow of energy, and black arrows show the flow of mass. All non-named mass flows are H_2, N_2, ammonia or any combination of the three.

Chemical equation [3]

$N_{2\,(g)} + 3H_{2\,(g)} \rightleftharpoons 2NH_{3\,(l)}$	[3]

3.4.5 Urea, Bosch-Meiser

The Bosch-Meiser method of producing urea is a well-established process for synthesizing urea. As shown in *Figure 8*, the main flows of mass are CO_2 and ammonia (NH_3), which are mixed in a high-pressure condenser to form ammonium carbamate (NH_2COONH_4). Which is moved to a high-pressure stripper, with high-pressure CO_2 as a stripping agent. Leaving a urea-rich liquid solution to be further processed in the reactor and separator. Gaseous water vapor can, at the final step, be used as thermal exchange while urea is slowly removed as a product ready mass (Bargiacchi, Antonelli and Desideri, 2019; Ding *et al.*, 2023). The rest of the processes are there to increase energy, material efficiency, or remove impurities. The model will incorporate these efficient matters while keeping the code streamlined. Bargiacchi, Antonelli and Desideri (2019) also produce a Sankey diagram showing some of the in- and outflows of energy stemming from the input value of 1 kilo-mol per second of H_2 as the basis. The study does incorporate some efficiencies, it is adjusted for a "single pass" through the production system, rather than having mass be recycled. making the impact extremely high if incorporated directly. Additional assumptions fill that gap, purge losses are assumed to be at 1% of fresh input from ammonia and CO_2, due to extensive recycling of feedstock (Soliman, 2019). Urea synthesis has a low single-pass rate of up to 10%, this is assumed to be the case for the study. To adjust energy usage fairly, energy usage before the loop (Compressors) is reduced to 10% of the article's energy usage. The resulting values were converted to per kg urea since the article produces the mass flow of urea as a product. The flows also specify the energy requirements for electricity, heat, and chemical feedstocks. The flows, Chemical equations [1] and [2], together with the functional unit, could calculate the desired inputs and outputs for the model in either kWh per kg urea or kg per kg urea, detailed in Appendix 1 (Bargiacchi, Antonelli and Desideri, 2019).

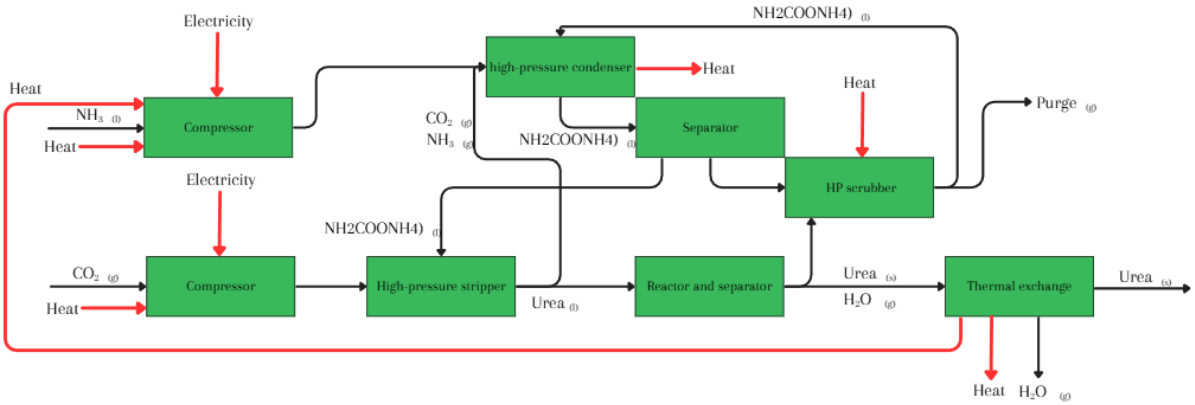


Figure 8. Internal flow of the Bosch-Meiser process, based on Bargiacchi, Antonelli and Desideri (2019). Red arrows show the flow of energy, and black arrows show the flow of mass. In some mass flow cases, such as the flow between the Separator and the High-pressure stripper, the named mass is only the main flow. Parts of the mass flow could be urea or CO_2 gas, as in the case of the aforementioned mass flow.

Chemical equations [1] and [2]

Ammonia and carbon dioxide to ammonium carbamate

$$2NH_{3\,(l)} + CO_{2\,(g)} \rightleftharpoons NH_2COONH_4 \qquad [1]$$

To urea and water

$$NH_2COONH_{4\,(l)} \rightleftharpoons CO(NH_2)_{2\,(s)} + H_2O_{\,(g)} \qquad [2]$$

3.5 Impact Assessment

To evaluate greenhouse gas emissions and energy use, a way to connect the flows with data was needed. To achieve that, the study used the tool SimaPro and the database Ecoinvent3. The database contains extensive data regarding industrial processes, materials, and energies. Which is used by SimaPro to calculate and facilitate the assessment. The following two methods were suitable to evaluate the production of urea and achieve the study´s objective. The first "IPCC 2021 GWP100 (incl. CO_2 uptake) V1.03" calculates net Global warming potential (GWP) in the unit kg CO_2 eq per optional metric. That metric was primarily kg for mass and kWh for energy. Only volume was an option for one occasion, which was later converted to kg. On another occasion, tonne-km was the best available metric, which was later "converted" to kg-1000km. The method was a clear choice to provide decision-makers with sufficient insights to evaluate potential climate impact. The second evaluation method was "Cumulative Energy Demand V1.12 / Cumulative energy demand", which calculates non-renewable primary energy and renewable primary energy per metric. The same metrics were used with this method as with the GWP method. The primary energies were expressed in MJ and converted to kWh. Primary energy and its separation into non-renewable and renewable options are useful for decision-makers to gain insight into sustainability. Each flow sent for evaluation had to be matched with a measured flow in Ecoinvent3, which, at times, may have been too generalized, as visible in Appendix 1.

4. Results and analysis

This chapter answers the research questions stated in the introduction. Research question 1 will be addressed in the section titled "Uncertainty analysis of urea production." Research question 2 will be answered under the Evaluation of urea production.

4.1 Potential factors and key uncertainties impacting urea production

The factors impacting urea production are endless, however, factors that have the potential to significantly impact greenhouse gas emissions or energy usage in urea production are presented below in Table 6.

Table 6. Area data: are uncertainties that can be observed and answered by empirical evidence from experiments or measurements. Area model: is a collection of uncertainties that can be found due to the use of mathematics, statistics, or logical conclusions drawn from observing input and output flows. This has no real impact on the results and was used when identifying them as uncertainties. Uncertainty types categorize uncertainties as standard and non-standard and were categorized as such depending on whether articles treat them as factors or if the author thought they were important enough to have them as factors whilst constructing the model. Non-standard uncertainties were identified during the uncertainty analysis and affected how some standard uncertainties were defined and calculated. Names of the uncertainties, an explanation, and thoughts on the uncertainties are presented under the table.

Uncertainty	***Area***	**Uncertainty types**	**Description**
1. *Compressor efficiency*	Data	Standard	Variations in compressor efficiency exist in the industry. Identified range 75% to 94%, with the study´s efficiency calculated at 85% (Valente *et al.*, 2020; García *et al.*, 2025). The factor would affect the electricity requirement of the compressor throughout the entire foreground system. It is not included in the factor-based analysis, but its impact is estimated below.
2. Technology pathways	Data	Standard	Different production technologies, detailed in 3.1.
3. Additional renewable input flows	Data	Standard	Detailed in 3.1.
4. C*arbon capture*	Data	Standard	Uses energy to capture CO_2 gas from purges exiting the foreground system, compresses it, and stores it. The range is from no CO_2 capture from gas leaving the foreground system to all CO_2 leaving the foreground system that emanates from purges. An estimation of the impact is conducted below.
5. *On-site IS*	Model	Standard	Detailed in 3.1.
6. *Urea transport to Sweden*	Model	Standard	The movement of a functional unit worth of urea, within the European borders, to an arbitrary Swedish position in Sweden. The factor impact is on the border of the scope foreground system. It is not included in the factor-based analysis, however, an extreme estimation is executed below.
7. *Non-incorporated mass flows*	Model	Standard	Several mass flows are too small for an individual impact assessment and are negligible. However, an estimation of their impact is conducted below.
8. *Electricity source*	Model	Standard	Investigate the impact of urea production using only the best and only the worst options in a geographical electricity mix on climate performance, measured by greenhouse gas emissions and primary energy use for electricity in Sweden or Europe. However, geographical location and *Additional renewable input flows* address the interesting aspects of this factor and were therefore not included in the factor-based analysis.
9. *Mass flow distance within the production system*	Model	Standard	Potential impact of separating the production of H_2, N_2, ammonia, or urea away from the production system. The factor would have affected the arrows between production technologies in *Figure 1.* It is a key factor, excluded from the factor-based analysis based on assumptions made in Section 3.3. The justification of the exclusions is discussed in Section 5.2.
10. *Ammonia utilization in urea production*	Data	Non-standard	Urea requires more or less ammonia due to different utilization variations, the variation is supposedly very low. However, it is interesting, since it cascades into the energy requirements

			for ammonia synthesis and the mass requirements for H_2 and N_2 production. The factor would have therefore affected the ammonia going into the process of urea synthesis and all that is before it in *Figure 1.* The factor is estimated below.
11. *H_2 capture efficiency in electrolysis*	Data	Non-standard	Electrolysis technology transforms water molecules into H_2 and O_2, capturing H_2 gas with sufficient purity means possible mass losses. The H_2 capture efficiency ranges widely, which can cause an estimated impact as described below.
12. *Methane transport leakage in the pipeline*	Data	Non-standard	Leakage variation of methane during pipeline transport from the methane refinery source to the urea production site. The factor would only affect methane going into the foreground system in *Figure 1.* Its potential impact is estimated below.
13. *Methane supply distance and renewability*	Data	Non-standard	Tries to capture all the ways methane can impact the urea systems' environmental performance, which is identified as distance and renewability. The factor would only affect methane going into the foreground system in *Figure 1*. The justification to exclude is discussed in Section 5.2.
14. *On-site O_2 utilization*	Model	Non-standard	Sees the possibility to positively impact greenhouse gas emissions and primary energy usage by directing the oxygen-rich purge gas from electrolysis. The factor would have necessitated and affected a new dashed mass flow arrow leaving the H_2 production in *Figure 1.* A feasible option is to use the oxygen on-site for flaring. Its potential impact is discussed in Section 5.2.
15. *Electricity fluctuations*	Model	Non-standard	The factor concerns fluctuations in the electricity mix due to seasonal changes and day cycles, and how adjusting operating times can improve environmental performance. Its exclusion and potential impact are discussed in Section 5.2.
16. *Production-size related efficiency*	Model	Non-standard	This factor was intended to analyze possible size advantages, such as efficiency improvement in certain processes. The factor could have affected any number of components in the foreground system shown in *Figure* 1. Its justification for exclusion is discussed in Section 5.2.

16 uncertainties were treated as potential key factors, including all standard uncertainties, which were thoroughly investigated. A large subset of the 16 uncertainties was also temporarily integrated in the initial model and compared to *Basic conditions*, i.e., the assumptions stated in the methodology without factor impact. The subset factors presented below was temporarily integrated, compared with *Basic conditions*, and ultimately excluded from the scenario due to lack of impact. Those were the uncertainties nr: 1, 6, 7, 10, 12, and 13.

Compressor efficiency (Uncertainty #1) has an identified industry variation range of 75% from Valente *et al.* (2020) to a theoretical maximum of 94% according to García *et al.* (2025). The model assumes 85% compressor efficiency. To analyze how efficiency variability affects the results, the compressor's electricity demand was adjusted using relative efficiency scaling factors. These factors are defined as the ratio of the actual efficiency used to the lowest identified efficiency and to the theoretical maximum.

The greatest impact under *Basic conditions* was for Europe, which saw a change in greenhouse gas emissions by 13% (+7.4 to -5.3) and primary energy usage change by 15% (+8.8 to -6.3). Swedish conditions have their greenhouse gas emissions altered significantly less, up to 2.5%, since the *Electricity source* has a higher share of *Additional renewable input flows*. It is therefore close to certain that the impact on greenhouse gas emissions will continue to shrink and be almost eliminated as the share of *Additional renewable input flows* increases. However, primary energy usage barely decreased and is still rounded up to 15% (+7.9 to -5.7) change. *Electrolysis*, which also had its impact estimated, had similar impact levels.

Carbon capture (Uncertainty #4) was integrated into the model for a time before it was chosen to highlight other factors. European conditions reduced greenhouse gas emissions by 10% compared to its *Basic conditions*. That reduction is greater, at 13% under Swedish conditions compared to its *Basic conditions*. This impact increased with *Additional renewable input flows* to approximately 40% for both Swedish and European conditions. *Production p*athway 2 (ELECTROLYSIS) was barely affected by the factor. The primary energy usage increases with *Carbon capture*, but to a negligible degree.

Urea transport to Sweden (Uncertainty #6) was estimated under the following assumptions: the distance was set at the rather extreme value of 3.500 km, corresponding to a journey from Lisbon to Linköping. The truck was a 14-20-tonne lorry using standard diesel fuel. The fuel consumption, road usage, and several other assumptions were automatically chosen using NTMCalc Basic 4.0 (NTMCalc Basic 4.0, 2025). It was, however, supplemented with diesel assessment data from Ecoinvent3 using “1 kg Diesel {RER}| market group for diesel” and the density of diesel at 0.845 kg/l (LearnMetrics, 2025). Based on the assumptions, approximately 550 kg CO_2 eq was emitted and 2,100 kWh of primary energy was used for travel. An upper limit impact of under 5.0% for both greenhouse gas emissions and primary energy usage was estimated, which would have slightly worsened the results under European conditions. The transport is estimated to be more prominent when other factors are implemented, however, the impact of the transportation can also be reduced with more renewable transportation options.

Non-incorporated mass flows (Uncertainty #7) is the only standard uncertainty that is verified not to be a key factor. The original assessment was that some mass flows would be too low to yield

impact. However, all disregarded mass flows together could impact urea production, hence the uncertainty was chosen as a potential key factor. The non-calculated flows identified under *Basic conditions* were: 0.15 kg of MDEA solvent (ammonia), 1.0 kg of methanation catalyst, 0.16 kg of catalyst material for steam methane reforming, and 0.12 kg of potassium hydroxide for hydrogen electrolysis. Which summed up reached 1.4 kg. Available data on the flows in Ecoinvent3 found that ammonia had 2.7 times the Global Warming Potential and energy demand, compared to the most common substance, aluminum oxide (prominent mass in the methanation catalyst). Ecoinvent3 data showed that ammonia has 2.7 times the greenhouse gas emissions and primary energy usage as aluminum oxide. A safer initial estimate was to use the higher-impact value, since the main purpose was to assess the potential impact. Therefore, the higher impact of ammonia got to represent the per kg impact of all the non-calculated input mass flows. 570 kg of ammonia was already calculated in the model, and adding the total mass of 1.4 kg of non-calculated flows, or a 0.24% increase in ammonia, would not affect the total impact of ammonia, not even by 1%. This point is strengthened by the next paragraph, which estimates the impact of ammonia variation. Note that the estimate is based only on what the used mass flows could cause, given their prior production impact. The estimate provides no indication of their importance in urea production.

*Ammonia utilization in urea production (*Uncertainty #10), Borsch-Meiser is a mature production technology with not a lot of efficiency gains left to improve (Bargiacchi, Antonelli and Desideri, 2019). However, to estimate how important that efficiency is for the whole production system, the rather extreme ammonia requirement range of ±10% was used to estimate the impact of different utilization levels. For the model, this meant the required ammonia amount was multiplied by 1.1 and 0.9 under *Basic conditions*. The maximum impact on greenhouse gases ± 1.7%, occurring under Swedish conditions, and a primary energy impact of ±1.9% in European conditions. The overall impact under *Basic conditions* is not negligible and is considered a key factor.

H_2 capture efficiency in electrolysis (Uncertainty #11) has a variation of 82% to 100% captured H_2 (García *et al.*, 2025). This had no effect on *Basic conditions*, since electrolysis is only used in *Electrolysis*. However, the potential impact on *Electrolysis* was considerable, since no definitive H_2 capture efficiency was stated in Valente *et al.* (2020). The total variation range of resources needed to produce the same amount of H_2 is then 40% (+22% to -18%), a significant range. The H_2 production requirement was therefore multiplied by 100/82 and 0.82 in the model under *Basic conditions*, apart from using *Electrolysis*. The results showed that primary energy usage was the most affected with a total impact range of 7.3% (+4.0% to -3.3%) for European conditions and 6.7% (+3.7% to -3.0%) for Swedish. Greenhouse gas emissions under European conditions were also significantly affected, with a total impact range of 5.80% (+3.5% to -2.3%), whereas they were 1.5% (+0.83% to -0.69%) under Swedish conditions.

Methane transport leakage in the pipeline (Uncertainty #12) uses assumptions detailed in Section 5.2. The assumed extreme distance of 6600 km, the estimated impact under *Basic conditions* is as follows. The maximum impact occurred under Swedish location, where greenhouse gas emissions increased by 4.2% and primary energy use by 5.5%. *Methane supply distance and renewability* (Uncertainty #13) were estimated roughly using methane leakage estimates from transportation data, together with methane sources detailed in Appendix 1. Its

maximum greenhouse gas impact at the maximum feasible distance used above, and using renewable biogas. The impact was an increase of 9.7% in Sweden and 4.8% in Europe. Primary energy usage had its greatest impact under conditions with no distance and no biogas. The impact was a decrease in primary energy usage of -4.8% for Swedish conditions and -3.9% for European conditions.

4.2 Evaluation of urea production

This section investigates Swedish urea production under different conditions and compares it with European urea production in terms of greenhouse gas emissions and primary energy use. The figures below illustrate the results of the factor-based scenario analysis, which used the most impactful factors, as estimated.

The results from the factor-based scenario analysis for greenhouse gas emissions are illustrated in *Figure 9*. A follow-up text analyzes the figure and the impact of factors.

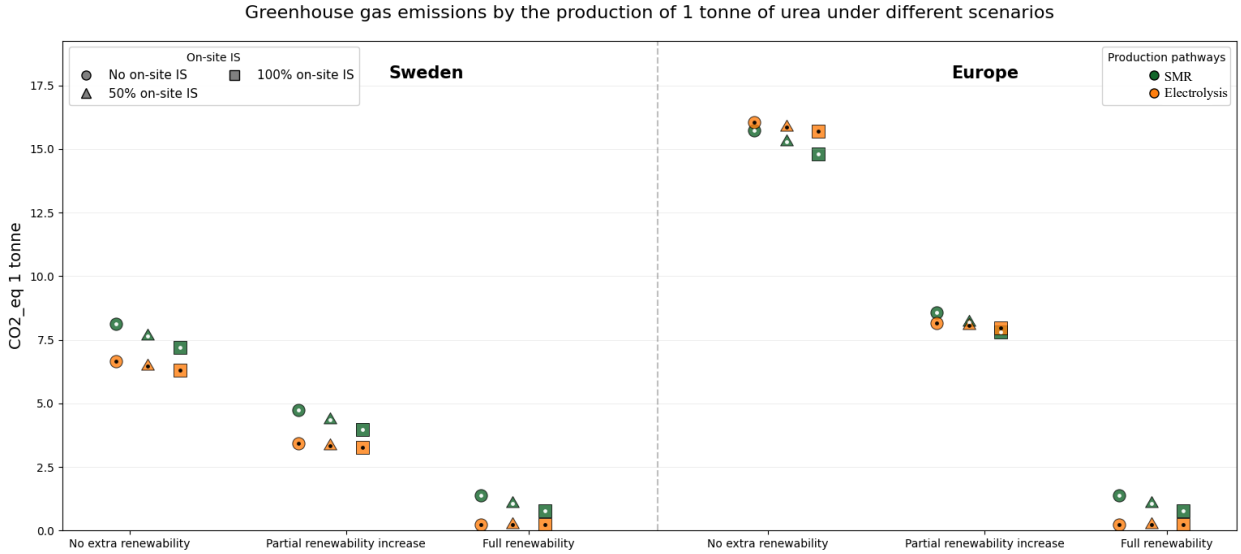


Figure 9. The grid shows the conditional data of greenhouse gas emissions for 1 tonne of urea produced in Sweden and for a generalized European location. The geographical locations are identified as the top horizontal labels. Production pathways are identifiable in the top-right corner, with SMR indicating Steam methane reforming hydrogen gas production and SMR indicating Water electrolysis hydrogen gas production. The level of IS is indicated by the shape. Where circle indicates no on-site IS activity, triangle 50% on-site IS, and the square 100% on-site IS. The figure is further divided along the horizontal axis, with the bottom smaller divisions representing Additional renewable input flows incorporated. No extra renewability means that the renewability is the normal one for that area, with reasonable but worse material used when a regional mix was not an option. Full renewability indicates that several input flows are replaced with a much more sustainable alternative for that geographical location, detailed in Appendix 1. Partial renewability is the assumption that half of the flows from Basic conditions are substituted for those of Full renewability.

Basic conditions (emissions without factor variation) are the first pair of circles to the far left of Swedish and European conditions. All numbers are emissions from the production of 1 tonne of urea. *Basic Swedish conditions* emit an estimated 8,100 kg CO_2 eq, whilst *European conditions* emit a maximum of 15,700 kg CO_2 eq. For *Electrolysis*, those greenhouse gas emissions are changed to 6,600 kg CO_2 eq under Swedish conditions and 16,000 kg CO_2 eq under European conditions. However, an estimate of the possible efficient industries could range from 200 to 15,000 kg CO_2 eq under Swedish conditions and from 200 to 32,000 kg CO_2 eq under European

conditions. The wide variations are primarily due to differences in electricity and heating mixes across Europe.

The results indicate that CO_2 emissions can be significantly reduced to approximately 230 kg CO_2 eq under Swedish conditions and 240 kg CO_2 eq under European conditions when using *Additional renewable input flows* and *Electrolysis*. The geological disparity is diminished through an increased share of *Additional renewable input flows.* For *SMRP* (the option used under *Basic conditions*), their greenhouse gas emissions can only be reduced to around 760 CO_2 eq under Swedish conditions and 780 CO_2 eq under European conditions, which also requires full implementation of *On-site industrial symbiosis (IS).*

Note that the most impactful factor for greenhouse gas emissions is *Additional renewable input flows. Geographical location* is the second most impactful factor, however, *Additional renewable input flows* decrease that impact. *Additional renewable input flows* also influence production pathways, by seemingly improving *Electrolysis*' performance more than *SMRP*'s. An interesting observation is that *Electrolysis* is worse than *SMRP* at low levels of *Additional renewable input flows* in the European context, but performs better as the share of *Additional renewable input flows* increases. The reason this is not the same in Sweden is that the national electricity mix is more renewable than the European regional electricity mix.

On-site IS has a limited impact on greenhouse gas emissions under *Basic conditions,* reducing emissions by 11% in Sweden and 5.9% in Europe. However, the largest relative impact occurs under *Full renewability,* in which case IS reduces *SMRP* emissions by 44%.

The results from the factor-based scenario analysis for primary energy usage are illustrated in *Figure 10*. A follow-up text analyzes the figure and compares it to *Figure 9*.

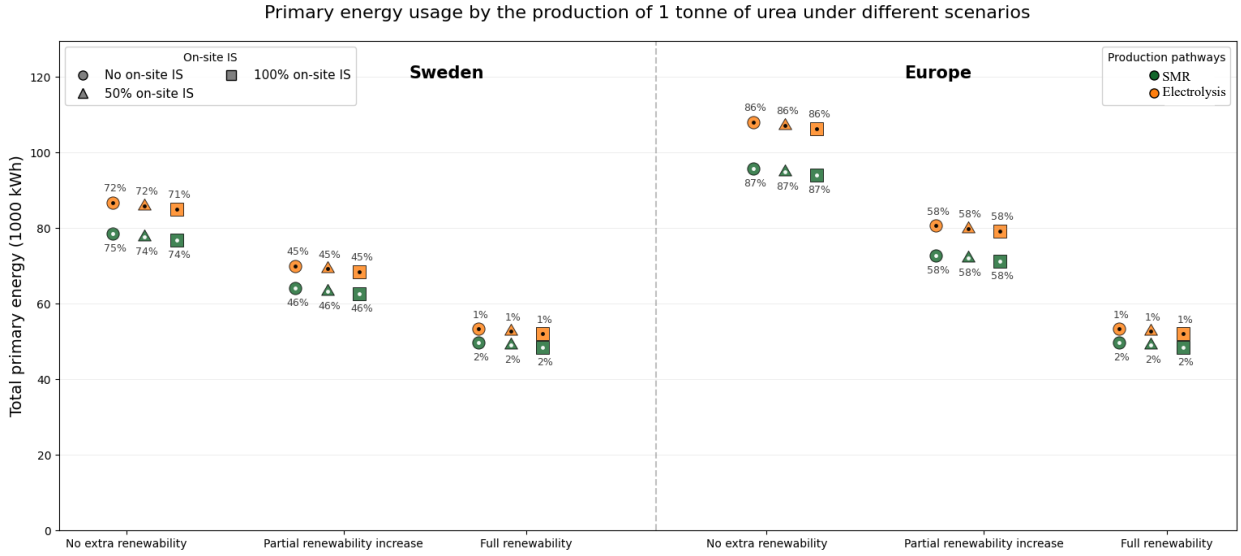


Figure 10. The grid shows the conditional data of primary energy usage for 1 tonne of urea produced in Sweden and for a generalized European location. The geographical locations are identified as the top horizontal labels. Production pathways are identifiable in the top-right corner, with SMRP indicating Steam methane reforming hydrogen gas production and SMRP indicating Water electrolysis hydrogen gas production. The level of IS is indicated by the shape. Where circle indicates no on-site IS activity, triangle 50% on-site IS, and the square 100% on-site IS. The figure is further divided along the horizontal axis, with the bottom smaller divisions representing Additional renewable input flows incorporated. No extra renewability means that the renewability is the normal one for that area, with reasonable but worse material opted when a regional mix was not an option.

Full renewability indicates that several input flows are replaced with a much more sustainable alternative for that geographical location, detailed in Appendix 1. Partial renewability is the assumption that half of the flows from Basic conditions are substituted for those of Full renewability.

Basic conditions are still represented by the colored circles to the left-most side of Sweden and Europe, respectively. All numbers represent the primary energy use in producing 1 tonne of urea ready for market. The primary energy usage is estimated to reach 78,000 kWh in Sweden and 97,000 kWh in Europe. Plausible range of primary energy usage in efficient industries, without these factors affecting them, could still range from under 40,000 kWh to 120,000 kWh in Swedish conditions, and from under 40,000 kWh to 170,000 kWh in European conditions. A broad-stroke observation is that primary energy cannot be reduced to the same extent as greenhouse gas emissions, however, factors still significantly affect primary energy.

Additional renewable input flows is the most impactful factor and is estimated to reduce the amount of primary energy used by 37% in Sweden and 48% in Europe. Under *Basic conditions*, Sweden uses 75% primary energy from non-renewable sources, and Europe 87% non-renewable primary energy. That changes with increased *Additional renewable input flows*, with almost complete renewability at *Full renewability*. No other factor affects the primary energy ratio to the same extent, although *Geographical location* does affect the ratio significantly under *Basic conditions.*

The *Geographical location* has a significant impact in most scenarios, with the largest difference under *Basic conditions*, where European conditions were calculated to use 21% more primary energy than Swedish. That indicates that geographical location is more important for primary energy than for greenhouse gas emissions performance.

The *Production pathway Electrolysis* has higher values in both *Geographical locations*; the difference diminishes as *Additional renewable input* flows increase, but it is not eliminated.

On-site IS has a limited impact on primary energy use under *Basic conditions*, as with greenhouse gas emissions. Their relative impact also increases under *Additional renewable input flows*, but by much less this time, with reductions of 2.1% under *Basic conditions* and 2.6% at *Full renewability.*

Overall, Swedish production conditions are simulated to perform significantly better than European conditions in most scenarios with respect to greenhouse gas emissions and primary energy usage.

5. Discussion

This chapter discusses key topics regarding results, validity of assumptions, methodology, and future research.

5.1 Results

Additional renewable input flows have an indisputable positive effect on reducing greenhouse gas emissions and primary energy usage. The consequential discussion, however, concerns whether the local benefits at the production site are equivalent to regional effects. Renewable supply in any region is finite, and allocation flows to urea production can displace renewable supply from elsewhere. At the same time, basic national economic principles regarding free markets dictate that increased demand will put pressure on supply. However, it does not dictate how much of the demand will actually be fulfilled by the supply. Furthermore, the results indicate that fully renewable energy and mass flows to the production site do not imply that the entire primary energy input becomes renewable. This is supported by the Ecoinvent3 data presented in Appendix 1, which confirms the presence of non-renewable resource traces in all flows. These non-renewable fractions can arise from several sources, such as fossil-based machinery in upstream material extractions, or operational infrastructure using regional electricity mixes to power hydropower facilities. This fact strengthens the fact that Swedish conditions are better at reducing the environmental footprint, since it already has a significant amount of renewable supply in its energy mix.

The two different production pathways, *SMRP* with Steam methane reforming and *Electrolysis* as the hydrogen gas (H_2) production technologies, are interesting factors to discuss. It shows that production technologies and production technology pathways should be chosen based on available input sources. From an environmental standpoint, this study indicates that a production plant close to a biogas facility and with a mediocre electricity supplier should use a production pathway like *SMRP*, which uses biogas, rather than a pathway like *Electrolysis*, which uses more electricity. Sweden, which has a generally good electricity mix compared to the rest of the world, should investigate Water electrolysis as a H_2 production method.

5.1.1 Validity of results

A comparison with Ecoinvent3 data to produce urea for the European market (RER) and the rest of the world (RoW) was conducted, since those are the only two applicable options in Ecoinvent3. Greenhouse gas emissions varied substantially between the two conditions, with RER emitting 1.3 tonnes of carbon dioxide equivalents (CO_2 eq) and RoW emitting 1.6 tonnes. This falls within the scenario range, which strengthens the study´s validity. Granted, most values are lower than in the average scenarios, as illustrated in *Figure 10.*

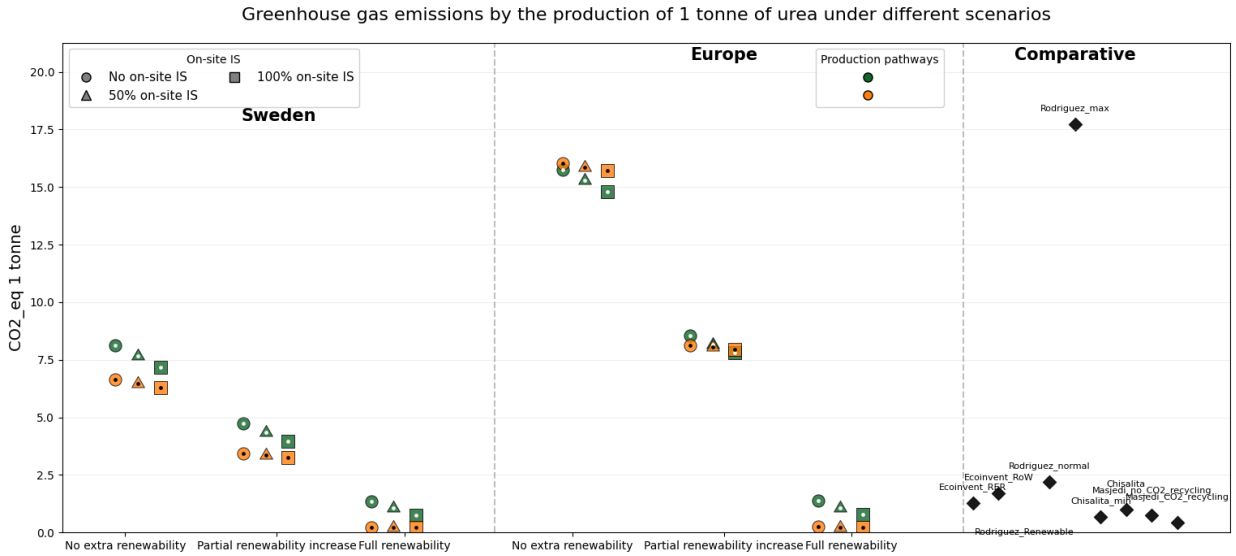


Figure 11. The grid shows the conditional data of greenhouse gas emissions for 1 tonne of urea produced in Sweden and for a generalized European location. The geographical locations are identified as the top horizontal labels. Production pathways are identifiable in the top-right corner, with SMRP indicating Steam methane reforming hydrogen gas production and SMRP indicating Water electrolysis hydrogen gas production. The level of IS is indicated by the shape. Where circle indicates no on-site IS activity, triangle 50% on-site IS, and the square 100% on-site IS. The figure is further divided along the horizontal axis, with the bottom smaller divisions representing Additional renewable input flows incorporated. No extra renewability means that the renewability is the normal one for that area, with reasonable but worse material opted when a regional mix was not an option. Full renewability indicates that several input flows are replaced with a much more sustainable alternative for that geographical location, detailed in Appendix 1. Partial renewability is the assumption that half of the flows from Basic conditions are substituted for those of Full renewability. The right-most graph shows values taken from other similar articles that calculate the climate impact for urea, all but (Chisalita et al, 2020) have been converted to use the same functional unit and a similar foreground system.

Ecoinvent3 also reports that urea production uses 9,800 kWh of primary energy under RER conditions and 10,000 kWh under RoW conditions. This considerably under the span of this study, as illustrated in *Figure 11*.

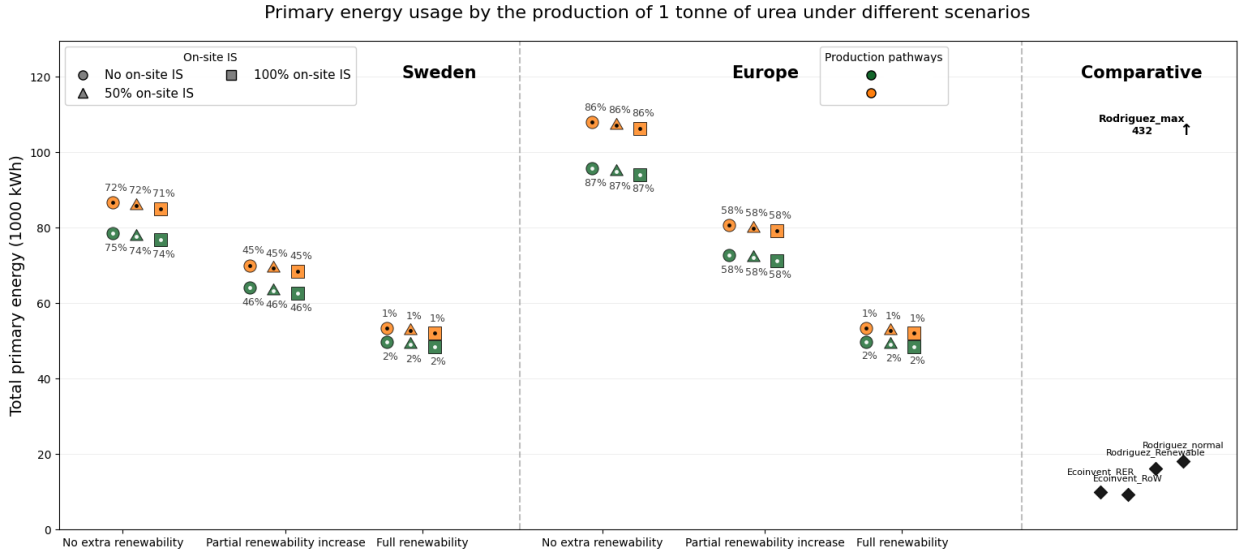


Figure 12. The grid shows the conditional data of greenhouse gas emissions for 1 tonne of urea produced in Sweden and for a generalized European location. The geographical locations are identified as the top horizontal labels. Production pathways are identifiable in the top-right corner, with SMRP indicating Steam methane reforming hydrogen gas production and SMRP indicating Water electrolysis hydrogen gas production. The level of IS is indicated by the shape. Where circle indicates no on-site

IS activity, triangle 50% on-site IS, and the square 100% on-site IS. The figure is further divided along the horizontal axis, with the bottom smaller divisions representing Additional renewable input flows incorporated. No extra renewability means that the renewability is the normal one for that area, with reasonable but worse material opted when a regional mix was not an option. Full renewability indicates that several input flows are replaced with a much more sustainable alternative for that geographical location, detailed in Appendix 1. Partial renewability is the assumption that half of the flows from Basic conditions are substituted for those of Full renewability. The right-most graph shows values taken from other similar articles that calculate the climate impact for urea, all but (Chisalita et al. 2020) have been converted to use the same functional unit and a similar foreground system.

A comparison to the precursor thesis by Rodriguez (2023) which is a similar study with a wider range of production pathways, more theoretical calculations, and a broader impact span. Rodriguez (2023) has one renewable production pathway, which is a carbon sink, absorbing 770 kg CO_2 eq and has its label barely visible in *Figure 10. This* study does not have a carbon sink, since material and energy is retained if possible, and Several results from Rodriguez (2023) are similar to this study in terms of greenhouse gas emissions, which strengthens this study. The study also has maximum values higher than *this* study in both greenhouse gas emissions and primary energy usage, which again strengthens the results of *this* study. Although it raises the questions if *this* study is too precise instead of accurate. Incorporating the factors *compressor efficiency* and *H_2 capture efficiency* would expand this study´s primary energy usage span. However, the value-span in Rodriguez (2023) would still encompass *this* study by a fair margin. One reasonable conclusion when looking at other studies is that Rodriguez (2023) has an excessive broadness, and does not reflect industry values, which then strengthens *this* study, since it is between the maximum and minimum values. No other strong conclusion can be drawn from this comparison.

A subsequent comparison with Chisalita *et al*. (2020) is performed to strengthen the study's validity. It is a study with a narrower foreground system that ends at ammonia and starts with materials entering the H_2 production at 900 °C and 27 bar. The omission of the required energy is significant, which is why this study does not omit it. Its value is presented without adjustments to the foreground system, and without adjustment of its functional unit, which is 1 tonne of ammonia. That is more ammonia than *this* study uses (0.6 tonne). The values presented in *Figure 10* are from two different production methods and are lower than the majority of *this* study´s values, which still supports the validity of this study, albeit marginally.

The last comparison made was to Masjedi *et al.* (2024), it conducted a LCA on an annual production of urea in Iran, the production methods for H_2 and nitrogen gas (N_2) were unspecified but included in the calculations. Masjedi et al. (2024) report that Iranian urea production emits 750 kg CO_2 eq per ton of urea produced, ranking second lowest among studies. However, the study does not loop mass flows observed in other sources and with the introduction of a CO_2 recovery system, the emissions are reduced to 450 kg per CO_2 eq emissions. It does not undermine the validity of this study, but it supports the argument that it has higher greenhouse gas emissions and uses more primary energy than most industries, theoretical or real.

The higher values can be explained by assumptions made when collecting mass- and energy flow data from Bargiacchi et al. (2019). *That* article calculates mass and energy flows for a "single pass" in the production system's steps. *That* article stated that it reflected industry practices, so the data were deemed adequate and incorporated. However, in another article and in industry practice, the flow of mass was much more efficient, with almost no loss of primary chemicals ammonia and CO_2 (Erfani, Baharudin and Watson, 2024). The mass flows from Bargiacchi *et al*.

(2019) were corrected by mass, however, the adjustment of energy was not enough or inaccurate. It, together with the Haber-Bosch data, is the origin of the higher estimation of impact. The possible implication of higher impact in energy, due to Haber-Bosch and Bosch-Maiser article presentation of flows, is that factors affecting primary energy sources would appear more influential. A prime example would be the *Geographical location*. On the other hand, factors not directly affecting production, such as *Transport of urea to the Swedish market* or factors related to methane leakage, would be presented as less significant. Additionally, there is a lack of factors focused on energy efficiency, such as compression efficiency and H_2 capture efficiency, which can lower primary energy use and narrow the gap between this study and most other data.

Environmental performance varies greatly for urea production, and within that wide variation, this study´s validity is adequate. It is situated within industry practices and countries that use greenhouse gas recovery systems and above theoretically perfect systems. The study might have a slightly askew factor impact, however, that should not significantly affect the primary conclusions.

5.2 Key assumptions and evaluation regarding factor filtering

The majority of the identified factors that had the potential to impact the environmental and energy performance of a urea production system were excluded from the factor-based scenario analysis. There is also an endless number of other unidentified factors that can affect the urea production system. Identifying which factors have the greatest impact on the system remains challenging, and the academic literature offers no clear consensus. Chisalita et al. (2020) examine a handful of factors and provide the closest approximation to a holistic overview. However, the criteria for factor selection are not fully discussed, and other potentially influential factors are not considered

Initial estimations based on logical reasoning and rapid estimations were, in some cases, insufficient. For example, non-calculated flows, a standard uncertainty, had significantly less impact than the non-standard uncertainty associated with H_2 capture efficiency. This raises the question of whether the initial identification of 16 factors may have excluded variables that could later emerge as key factors. The solution cannot be to include all possible factors, since each additional factor requires well-founded assumptions in order to be properly estimated, which is both time- and resource-intensive. Likewise, the factor-based scenario analysis cannot use all identified factors in the model since the number of scenarios and assumptions increases rapidly under a factorial scenario-building structure. This presents a challenge for anyone attempting to replicate a similar study, involving a factor-based analysis on a complex production system with several production technologies.

5.2.1 Excluded standard uncertainties

The selected *Compressor efficiency* (Uncertainty #1) range is considered robust. García *et al.* (2025) present lower efficiencies, however, these values correspond to disproportionate mass flows and compressor start-up conditions. This is not the condition this study focuses on; it examines extreme yet feasible continuous industrial flows. Which is why the lowest compressor efficiency mentioned in other literature was used. The impact on greenhouse gas emissions is assumed to be reduced with the introduction of *Additional renewable input flows*. Primary energy usage, on the other hand, could both increase or decrease depending on the influence of

other factors and therefore be particularly interesting as the next factor included in future factor-based analyses.

Carbon capture (Uncertainty #4) provides value in reducing greenhouse gas emissions. However, it primarily benefits industries with a significant CO_2 surplus escaping from their production system. This is an unlikely condition for urea production, since CO_2 is heavily recycled, and rightly so, given that the gas not only avoids pollution but also offsets the need to purchase new CO_2 for the system. The factor would have taken CO_2 from all production purges by capturing and storing the gas, using 0.0758 kWh of electricity per kilogram of captured gas to do so (Chisalita *et al.,* 2020). Less than 1% of the captured CO_2 escapes the process, and the model factor treats this as negligible (Chisalita *et al.,* 2020). *Carbon capture* was put in the factor-based scenario analysis with the factor parameters 'no-', 'half-', and 'full-carbon capture options, corresponding to the amount of CO_2 gas from purges that is being captured. The impact the factor had was estimated to be similar to *on-site IS,* with 4% reduction in greenhouse gas emissions under otherwise *Basic conditions*, which increased with *Full renewability* to 37% in Sweden and 48% in Europe. Which is highly significant. Overlap with *on-site IS* was the reason for its exclusion in the factor-based scenario analysis. On-site IS also offsets CO_2 from purges, but it recycles it rather than storing it. This offsets the demand for industry-grade CO_2 required as a material input. This makes on-site IS for CO_2 the more attractive option from an environmental performance perspective.

Mass flow distance within the production system (Uncertainty #9) is a key factor that is not estimated in the study, for reasons detailed in Section 3.3. However, that might change in the near future once the European Union´s REPowerEU initiative is realized. The initiative has set the target to produce 10 million tonnes of domestic (European) renewable hydrogen production, as well as importing another 10 million tonnes by 2030 (entsog *et al.*, 2025). The initiative also includes expanding the European H_2 grid by building new pipeline infrastructure and converting old natural gas infrastructure for H_2 transportation. In this context, Europe will significantly expand its network, while Sweden will build its first H_2 pipelines in the northeast of Sweden and one connecting to Gävle, both originating in Finland and connecting Sweden to the larger European grid. This development could change the availability and feasibility of sourcing ready-made H_2 for urea production, thereby avoiding the need for local production or incurring expensive transportation costs. The implementation of already-made H_2 would facilitate urea production in Sweden and in Europe. Renewable hydrogen would largely be dependent on the electricity source, making it unlikely that Swedish production would have improved environmental performance. However, that is not always the case for the rest of Europe.

Electricity source (Uncertainty #8) is a key factor, evident across several other factors, with an impact comparable to that of *Additional renewable input flows* and *Geographical location*. The factor would be a lesser version of what *Additional renewable input flows* and their lack of new insights were the reasons it was not included in the factor-based analysis. Changing only the electricity source to its best-performing renewable source, which is included in *Additional renewable input flows,* resulted in: a reduction to less than half of greenhouse emissions under original European conditions, and a reduction of primary energy quite similar to *Additional renewable input flow* illustrated in Figure *9. Electricity fluctuations (Uncertainty #15)* were excluded for the same reason, and because they have a lower impact, as the difference in average

mix is significantly less than the fluctuation in electricity. Seasonal variation in Swedish electricity ranges from 63% to 70% renewable flows (Statistics Sweden, 2026), while daily cycles were excluded due to the continuous nature of urea productions (Man and Ma, 2025). The impact would be less than a fifth of *Electricity source*, which is still significant under normal conditions.

5.2.2 Non-standard uncertainties

These are the factors that only had rough estimations conducted to determine if further and better estimations should be conducted, there could be further arguments for the exclusion. The first factor, Methane transport leakage in the pipeline (Uncertainty #12), refers to the natural gas (NG) transportation from the largest supplier to the European market, at least under *Basic conditions*. This transportation results in an average of 0.37 kg CO2 eq and requires 9.3 kWh, almost exclusively nonrenewable primary energy, per kg of methane produced. Furthermore, transporting 1 kg of methane 1000 km emits 19% of the greenhouse gases and 3.3% of the energy required to produce it. Transportation distances can be significantly longer than 1000 km, particularly for supplies originating in Algeria, Russia, and the United States, which are among the largest NG suppliers to Europe (Council of the European Union, 2024). A hypothetical pipeline length was therefore estimated as the direct distance from Palmer, Alaska, United States, to Linköping, spanning the Arctic Ocean and totaling 6600 km.

Under *Basic conditions*, the environmental impact from methane accounts for 144 CO_2 eq and 3650 kWh using *Electrolysis* and 320 CO_2 eq and 4400 kWh using the steam methane reforming pathway (*SMRP*), with the extreme distance added. When compared to Swedish and European *SMRP* conditions, these differences explain the impacts observed in section 4.1. This shows that even an extreme distance will not significantly change results from *Basic conditions*, it is worth noting that the impact could also decrease with assumptions of shorter distances, since the average distance from the largest suppliers to Europe is included. The same applies to biogas, but to a lesser extent. Additionally, the impact becomes more important if greenhouse gas emissions are reduced in the rest of the production, since this factor´s emission increase remains largely unaffected by other factors. It is therefore possible that this is a key factor, which would be apparent if used in the factor-based scenario analysis. The factor could change some of the values in conclusion, however, primary conclusions should still remain. The same applies to uncertainty number 13, despite its higher impact from using renewable biogas, which neutralizes the impact from *Additional renewable input flows* currently used in the methane transition impact in its factor. Separating *Additional renewable input flows* into *Electricity source* and *Methane impact* could have been interesting and is recommended for future studies.

On-site O_2 utilization (Uncertainty #14) is a non-standard uncertainty that warrants mention. The flow was treated as waste initially, waste that would not impact the environment or energy, which was why it was neglected. It was not until the main uncertainty analysis that the gas was further investigated, and the possible uses became available. The primary use was flaring, which could improve combustion efficiency. The effect is extremely limited, but since the mass O_2 flow is quite substantial, at a weight ratio of 9:1 of H_2 produced (Chisalita et al., 2020). It would not impact on the total environmental performance enough to warrant calculating, however, it could potentially have improved primary energy usage and greenhouse gas emissions enough to

improve the standing of *Electrolysis* compared to *SMRP*, since they are close in a majority of the scenarios. Despite this, the primary conclusions remain intact.

Production-size related efficiency (Uncertainty #16) is the last uncertainty to be investigated. Large-scale production enables optimization of operating conditions, leading to a substantial reduction in energy and material use per unit produced (Česká Společnost Chemického Inženýrství, 2006). Those operating conditions can be seen in the discussion and results in the form of *On-site IS* or *Carbon capture*. However, the factor only facilitates the implementation of those factors; it does not actually implement them. The identified uncertainty was therefore deemed too vague to be calculated. That could change if monetary factors are introduced, and the budget determines how many additional efficiency measures can be implemented. Nevertheless, it is redundant in this study and would not change the primary conclusions.

5.3 Validity of methodology

A comparison with individual European producers, as well as with specific countries such as Russia or Belarus, both major NBMF suppliers to the European market, including Sweden (RI.SE, 2023) could have been relevant. However, Europe as a region remains the most suitable comparison, since most NBMF imported to Sweden originate within the broader European market rather than a single country. In addition, primarily the conclusions would likely remain unchanged and widen the gap between geographical locations.

The chosen research philosophy supported the data collection process, contributing to the selection of higher sources in the evidence hierarchy. However, it remains unclear whether adopting a different philosophy, or using no philosophy, would have enabled faster data collection or a broader set of production pathways without compromising data security. Few studies reported data for identical production systems, scopes, and focal parameters, which limited the possibility of combining sources and instead necessitated identifying individual studies capable of providing all required data. Although this approach extended the duration of the study, it represents a methodological improvement compared to the preceding work on which this study builds, which relied extensively on calculated values and assumptions of idealized systems (Rodriguez, 2023). The methodology of refining keywords across databases was employed, as each search informed subsequent iterations. While consistent keywords would improve transparency slightly, this approach was less effective. The author therefore maintains that the benefits of iterative refinement outweigh the minimal loss of transparency.

The two pathways examined in the life cycle inventory represent fundamentally different H_2 production routes. One uses electricity as the primary input, while the other uses fossil gases and reaction-driven processes. Additional pathways involving other N_2 and ammonia-production technologies were available for inclusion in this study. However, they were not included due to procurement difficulties described in Section 3.1. These challenges were particularly evident for the urea production data. Which had its industrial values identified only for “single pass”. This means that material recirculation was not considered, requiring several assumptions about industry-standard recycling practices and their impact on energy use. It was not quite the same for ammonia production, however, the mass flow of H_2 required for the process is likely around 40% lower than calculated. Industries would have recycled that using pressure swing adsorption technology or a similar recovery technology, which uses electricity for the compressor but is still

more energy efficient than producing more H_2 with electrolysis and steam methane reforming technologies (Erfani, Baharudin and Watson, 2024).

This should not lower the study's accuracy, since *On-site IS* partially compensates for inefficient flows. Rodriguez (2023) addressed the challenge of good data by calculating it using chemical data, however, that would damage the purpose of the study, since flows and energies are not in a perfect system.

Each production technology obtained its input and output data from peer-reviewed literature, except for the mentioned urea assumptions and the energy input to produce H_2 using the steam methane reforming technology. These calculations are described in Appendix 3, which primarily uses perfect systems but also incorporates the more realistic element of compressor efficiency.

Primary methodological discussion arises from the use of standard and non-standard uncertainty classifications. These uncertainties could alternatively have been categorized as the more commonly used epistemic or aleatory classifications. Sorting them into uncertainties due to limited knowledge or inherent variability in natural phenomena (Li *et al.*, 2026). Epistemic uncertainties can be reduced through improved knowledge, whereas aleatory uncertainties remain unchanged regardless of new information. That classification has its uses, and recipients might believe it would better serve the study. However, that is not necessarily the case. The standard/non-standard classification better supported this study by enabling rapid yet accurate decision-making to determine which factors to calculate. It also facilitated the construction of the model, as the standard uncertainties were partially identified in tandem with its creation, strengthening both components.

The exclusion of certain standard uncertainties was primarily due to their negligible impact on total greenhouse emissions and primary energy usage. Nevertheless, they remain relevant for real-world urea production planning, as each factor´s contribution accumulates to influence the total performance, such as compressor efficiency, which had the highest estimated impact. Defining an appropriate system boundary was challenging, and each exclusion of a standard uncertainty required careful consideration.

If resources were unlimited and the author could iterate the factor integration process, then the factors would be as follows: *Compressor efficiency,* Technological pathways*, Carbon capture, Additional renewable energy input flows,* On-site IS, *Urea transport to Sweden, H_2 capture efficiency in electrolysis, Methane supply distance and renewability, and O_2 utilization.* The reason for *Compressor efficiency* would assess how energy performance, and particularly compressor quality, influences urea production. *Carbon capture* has previously shown significant effects on greenhouse gas emissions, and its interaction with *On-site IS* warrants further analysis. Including *H_2 capture efficiency* would more accurately identify how the *Electrolysis* pathway could impact primary energy consumption, making the differences between *Technological pathways* more nuanced. This study is currently dominated by *Additional renewable input flows,* but separating the factor into *Additional renewable energy input flows* and *Methane distance and renewability* would make it clearer to decision-makers which input flow is more impactful: energy or NG. In addition, the new factors would be impacted differently depending on other factors, for example *On-site IS*, would probably impact *Methane distance*

and renewability more. Finally, utilizing O_2 as a by-product could offset part of the electricity demand in the electrolysis production process and improve overall system efficiency.

The structure described in Step A of the Design of Experiments (DoE) is well-suited for factor manipulation and for a sensitivity analysis, which was the goal of this study. However, the implementation was only partly successful. Due to practical constraints, several factors in the model affected multiple scattered parts of the code rather than being listed systematically, as implied in Esguerra *et al.* (2021). Despite this, the model is soundly and logically structured. It is simply not optimized for direct manipulation of factors. This limitation does not alter the results, and the primary conclusions remain valid. The scenario graph also compares favorably with figures presented using a proper DoE structure, reinforcing the claim that the model is functional (Esguerra *et al.*, 2021). Nevertheless, a more accurate model structure could have made factor integration easier and allowed time for lower-priority tasks to be executed, leading to more robust outcomes.

Another contention regarding the model structure arises from the deviation in DoE, as shown in Step C, by not using a computerized function, such as a Global sensitivity analysis or a trade-off analysis, to analyze the results (Sauve *et al.*, 2023). While it is true that a well-functioning computer analysis can identify subtle correlations that a person cannot, it can also present patterns where none exist. In addition, computer analysis is time-consuming to build, and it risks leaving the person responsible over-reliant on it and unable to properly understand the study´s findings. The deviation from the analytic process is therefore a justified option.

5.4 Future research

There is a data vacuum regarding the general factor impact on Urea production, not just in the Swedish context, but across the industry. This study fills some of that vacuum; however, several academic articles can be made before the data vacuum is filled, since the production methods are plentiful and the possible factors are endless. This is a key takeaway from this study.

There is the possibility to expand *this* study, a lot of groundwork in making the assumptions has been performed and filling in with more data from pathways, even less established ones, would be interesting. The study´s objective would then be to create a robust comparison of production pathways. Alternatively, this study can be complemented by Material Flow Cost Accounting, enabling a more holistic trade-off analysis that helps decision-makers create effective economic policy incentives for companies to begin producing NBMF in Sweden.

A logical research study would be to identify suitable geographic locations for building urea production facilities. This is done by examining the critical factors identified in this study and evaluating local renewable potential, infrastructure, and other key factors for urea production. For example, sharing location with a biogas producer, for ready access to low-impact methane and renewable heat. Alternatively, placement near energy-intensive industries may facilitate heat, gas, and oxygen exchange for flaring.

The introduction of H_2 pipelines should be of interest. The optimal size for H_2 production in Sweden could change with the presence of a nearby H_2 pipeline. The impact on the European H_2 pipeline network from flooding it with renewable H_2 from electrolysis production technology

and a renewable electricity mix would also be an interesting subject. It would aid the European Union's goal of carbon neutrality, and possibly Sweden´s as well, if counted as a carbon offset.

The author believes that actual NBMF production will soon be operational in Sweden (RI.SE, 2023, 2025). A future research area could be conducted within a few years, focusing on a comparative analysis of NBMF production sites in Sweden to identify best practices. Other policy-related topics could be explored, such as carbon pricing on urea production or renewable energy certificates.

6. Conclusion

This study set out to evaluate whether domestic urea production in Sweden could offer environmental advantages compared to continued reliance on imported mineral fertilizers. Using a Life Cycle Assessment combined with a factor-based scenario analysis, 36 production scenarios were modeled to examine how different factors influence greenhouse gas emissions and primary energy use. The results show that among the analyzed factors, *Additional renewable input flows* proved to be the most influential, reducing greenhouse gas emissions by approximately 83% under Swedish conditions and 91% under European conditions, while also lowering primary energy use by 37% and 48%, respectively. These findings demonstrate that integrating renewable energy sources is a decisive factor in reducing the environmental footprint of nitrogen fertilizer production. The analysis also highlights the importance of *Geographical location*. In around two-thirds of the analyzed scenarios, urea production under Swedish conditions resulted in approximately 50% (±10%) lower greenhouse gas emissions than under European conditions. This is largely explained by Sweden's renewable electricity mix. However, the results also indicate that the environmental advantage of Swedish production decreases as renewable inputs are increasingly implemented across both regions. This suggests that technological improvements and the integration of renewable inputs can significantly reduce environmental impacts regardless of production location.

Overall, the findings suggest that establishing domestic urea production in Sweden could provide measurable environmental benefits, particularly when renewable energy and low-carbon production pathways are utilized. At the same time, the study demonstrates that technological choices, such as hydrogen production methods and energy sourcing, are equally important determinants of environmental performance. By identifying the factors with the greatest influence on emissions and energy use, this research contributes to a clearer understanding of how nitrogen fertilizer production can be made more sustainable and provides relevant insights for policymakers, industry actors, and future research on low-carbon fertilizer systems, while also highlighting the value of further studies that incorporate more detailed operational data and expanded environmental indicators to refine these assessments.

7. Bibliography

Abdoos, B., Di Benedetto, A. and Russo, D. (2025) "Integrated design, scale-up, and surrogate-based optimization of PSA systems for hydrogen purification from ammonia decomposition," *Chemical Engineering Journal*, 526, p. 170994. Available at: https://doi.org/10.1016/j.cej.2025.170994.

Ammenberg, J. (2023) *Miljömanagement: miljö- och hållbarhetsarbete i företag och andra organisationer*. Tredje upplagan. Lund: Studentlitteratur.

Argus Media (2024) "The history of fertilizers," *13 of August 2024*, 13 August, p. 36.

Ayres, R.U. and Ayres, L.W. (eds.) (2002) *A Handbook of Industrial Ecology*. Edward Elgar Publishing. Available at: https://doi.org/10.4337/9781843765479.

Bargiacchi, E., Antonelli, M. and Desideri, U. (2019) "A comparative assessment of Power-to-Fuel production pathways," *Energy*, 183, pp. 1253–1265. Available at: https://doi.org/10.1016/j.energy.2019.06.149.

Batstone, D.J. (2013) "Teaching uncertainty propagation as a core component in process engineering statistics," *Education for Chemical Engineers*, 8(4), pp. e132–e139. Available at: https://doi.org/10.1016/j.ece.2013.10.003.

Česká Společnost Chemického Inženýrství (ed.) (2006) *CHISA 2006, 17th International Congress of Chemical and Process Engineering, 27 - 31 August 2006, Praha, Czech Republic: CD-ROM of full texts*. *CHISA*, Praha (EFCE event, 663).

Chisalita, D.-A., Petrescu, L. and Cormos, C.-C. (2020) "Environmental evaluation of european ammonia production considering various hydrogen supply chains," *Renewable and Sustainable Energy Reviews*, 130, p. 109964. Available at: https://doi.org/10.1016/j.rser.2020.109964.

Council of the European Union (2024) *Where does the EU's gas come from?*, *consilum.europa.eu*. Available at: https://www.consilium.europa.eu/en/infographics/where-does-the-eu-s-gas-come-from/ (Accessed: November 12, 2025).

Devault, M., Woolf, D. and Lehmann, J. (2024) "Nutrient recycling potential of excreta for global crop and grassland production," *Nature Sustainability*, 8(1), pp. 99–111. Available at: https://doi.org/10.1038/s41893-024-01467-8.

Ding, J. *et al.* (2023) "Direct synthesis of urea from carbon dioxide and ammonia," *Nature Communications*, 14(1), p. 4586. Available at: https://doi.org/10.1038/s41467-023-40351-5.

EFSA Scientific Committee *et al.* (2018) "Guidance on Uncertainty Analysis in Scientific Assessments," *EFSA Journal*, 16(1). Available at: https://doi.org/10.2903/j.efsa.2018.5123.

Ekvall, T. *et al.* (2025) "Allocation in ISO, GHG Protocol, and Environmental Footprints." *SETAC-Europe Annual Meeting*, Vienna: Swedish life cycle center. Available at:

https://www.lifecyclecenter.se/wp-content/uploads/Ekvall-et-al-2025-Allocation-in-ISO-GHGP-and-EF-Presentation.pdf (Accessed: September 12, 2025).

Energimyndigheten (2022) *Sveriges import av naturgas*, *energimyndigheten.se*. Available at: https://www.energimyndigheten.se/nyhetsarkiv/2022/sveriges-import-av-naturgas/ (Accessed: January 14, 2026).

Engineering ToolBox (2025a) *Gases - Ratios of Specific Heat*, *engineeringtoolbox.com*. Available at: https://www.engineeringtoolbox.com/specific-heat-ratio-d_608.html (Accessed: October 14, 2025).

Engineering ToolBox (2025b) *Individual & Universal Gas Constants: Definitions, Values, and Applications*, *engineeringtoolbox.com*. Available at: https://www.engineeringtoolbox.com/individual-universal-gas-constant-d_588.html (Accessed: October 14, 2025).

Engineering ToolBox (2025c) *Methane Gas - Specific Heat vs. Temperature*, *engineeringtoolbox.com*. Available at: https://www.engineeringtoolbox.com/methane-d_980.html (Accessed: October 13, 2025).

Engineering ToolBox (2025d) *Nitrogen (N_2) Density & Specific Weight: Temperature & Pressure Data and Calculator*, *engineeringtoolbox.com*. Available at: https://www.engineeringtoolbox.com/nitrogen-N2-density-specific-weight-temperature-pressure-d_2039.html (Accessed: October 17, 2025).

Engineering ToolBox (2025e) *Properties of Saturated Steam - SI Units*, *engineeringtoolbox.com*. Available at: https://www.engineeringtoolbox.com/saturated-steam-properties-d_101.html (Accessed: October 13, 2025).

Engineering ToolBox (2025f) *Water Vapor - Specific Heat vs. Temperature*, *engineeringtoolbox.com*. Available at: https://www.engineeringtoolbox.com/water-vapor-d_979.html (Accessed: October 13, 2025).

entsog *et al.* (2025) *Hydrogen infrastructure map, showcasing concrete European hydrogen infrastructure projects and possibilities for transport routes and corridors*, *h2inframap.eu*. Available at: https://www.h2inframap.eu/ (Accessed: January 21, 2026).

Erfani, N., Baharudin, L. and Watson, M. (2024) "Recent advances and intensifications in Haber-Bosch ammonia synthesis process," *Chemical Engineering and Processing - Process Intensification*, 204, p. 109962. Available at: https://doi.org/10.1016/j.cep.2024.109962.

Esguerra, J.L. *et al.* (2021) "Landfill mining in Europe: Assessing the economic potential of value creation from generated combustibles and fines residue," *Waste Management*, 126, pp. 221–230. Available at: https://doi.org/10.1016/j.wasman.2021.03.013.

Feiz, R. *et al.* (2022) "The biogas yield, climate impact, energy balance, nutrient recovery, and resource cost of biogas production from household food waste—A comparison of multiple cases

from Sweden," *Journal of Cleaner Production*, 378, p. 134536. Available at: https://doi.org/10.1016/j.jclepro.2022.134536.

García, G.J.B. *et al.* (2025) "Energy Optimization of Compressed Air Systems with Screw Compressors Under Variable Load Conditions," *Mathematical and Computational Applications*, 30(5), p. 107. Available at: https://doi.org/10.3390/mca30050107.

Gas infrastructure Europe (2025) *European Biomethane Map*, *gie.eu*. Available at: https://www.gie.eu/publications/maps/european-biomethane-map/ (Accessed: January 14, 2026).

IEA (2021) *Ammonia Technology Roadmap*. Paris: IEI, The international Energy Agency. Available at: https://www.iea.org/reports/ammonia-technology-roadmap#overview (Accessed: September 22, 2025).

Incer-Valverde, J. *et al.* (2023) "Evaluation of the large-scale hydrogen supply chain and perspectives on LH2 regasification cogeneration systems," *Gas Science and Engineering*, 115, p. 205005. Available at: https://doi.org/10.1016/j.jgsce.2023.205005.

ISO (2011) *ISO 14051:2011(en), Environmental management — Material flow cost accounting — General framework*, *ISO*. Available at: https://www.iso.org/obp/ui/en/#iso:std:iso:14051:ed-1:v1:en (Accessed: October 8, 2025).

Jüntgen, H., Knoblauch, K. and Harder, K. (1981) "Carbon molecular sieves: production from coal and application in gas separation," *Fuel*, 60(9), pp. 817–822. Available at: https://doi.org/10.1016/0016-2361(81)90144-7.

LearnMetrics (2025) *Diesel Fuel Density In kg/m3, lb/gal, lb/ft3, g/cm3, kg/L (+ Chart)*, *leanmetrics.com*. Available at: https://learnmetrics.com/diesel-fuel-density-in-kg-m3-lb-gal-lb-ft3-g-cm3-kg-l-chart/ (Accessed: December 12, 2025).

Lee, S. *et al.* (2021) "Scenario-Based Techno-Economic Analysis of Steam Methane Reforming Process for Hydrogen Production," *Applied Sciences*, 11(13), p. 6021. Available at: https://doi.org/10.3390/app11136021.

Lewis, Robert, I., Svetlana (2012) "Producing nitrogen via pressure swing adsorption," *Chemical Engeneering Progress*, June 2012(108), pp. 38–42.

Li, P.-P. *et al.* (2026) "Aleatory and epistemic uncertainty in reliability analysis: An engineering perspective," *Structural Safety*, 119, p. 102666. Available at: https://doi.org/10.1016/j.strusafe.2025.102666.

Lifset, R. and Graedel, T.E. (2002) "Industrial ecology: goals and definitions," in R.U. Ayres and L.W. Ayres (eds.) *A Handbook of Industrial Ecology*. Edward Elgar Publishing. Available at: https://doi.org/10.4337/9781843765479.00009.

Liu, H.B., Jiang, C. and Xiao, Z. (2020) "Efficient uncertainty propagation for parameterized p-box using sparse-decomposition-based polynomial chaos expansion," *Mechanical Systems and Signal Processing*, 138, p. 106589. Available at: https://doi.org/10.1016/j.ymssp.2019.106589.

Lustenberger, P. *et al.* (2019) “Assessing the Performance of the European Natural Gas Network for Selected Supply Disruption Scenarios Using Open-Source Information,” *Energies*, 12(24), p. 4685. Available at: https://doi.org/10.3390/en12244685.

Man, J. and Ma, T. (2025) “Comparative techno-economic analysis of hydrogen supply in China: Green hydrogen vs green electricity,” *Journal of Renewable and Sustainable Energy*, 17(3), p. 034701. Available at: https://doi.org/10.1063/5.0259105.

Marschner, H. and Marschner, P. (2012) *Marschner’s mineral nutrition of higher plants*. 3rd ed. London Waltham, MA: Academic Press.

Martin, M. (2018) *Environmental assessment of the Sotenäs Industrial Symbiosis Network*. Governmental No. C 275. Svenska Miljöinstitutet, p. 38. Available at: https://ivl.diva-portal.org/smash/record.jsf?pid=diva2%3A1549675&dswid=-4555 (Accessed: October 2, 2025).

Masjedi, S.K. *et al.* (2024) “Urea production: An absolute environmental sustainability assessment,” *Science of The Total Environment*, 908, p. 168225. Available at: https://doi.org/10.1016/j.scitotenv.2023.168225.

Metson, G.S. *et al.* (2022) “Swedish food system transformations: Rethinking biogas transport logistics to adapt to localized agriculture,” *Sustainable Production and Consumption*, 29, pp. 370–386. Available at: https://doi.org/10.1016/j.spc.2021.10.019.

NTMCalc Basic 4.0 (2025) *Environmental Performance Calculator*, *Transportmeasures.org*. Available at: https://www.transportmeasures.org/ntmcalc/v4/basic/index.html#/ (Accessed: November 17, 2025).

O’Neill industrial (2020) *Nitrogen Generators*, *oneillcompressedair.com*. Available at: https://www.oneillcompressedair.com/nitrogen-generators/ (Accessed: December 23, 2025).

Peggy Shannon-Baker (2023) “Philosophical underpinnings of mixed research in education,” 17(4), pp. 280–389. Available at: https://doi.org/10.1177/15586898231201.

Reap, J. *et al.* (2008) “A survey of unresolved problems in life cycle assessment: Part 1: goal and scope and inventory analysis,” *The International Journal of Life Cycle Assessment*, 13(4), pp. 290–300. Available at: https://doi.org/10.1007/s11367-008-0008-x.

RI.SE (2023) *Jordbruksverket har kartlagt potential för inhemsk gödselproduktion*, *RI.se*. Available at: https://www.ri.se/sv/svenskanaringsplattformen/nyheter/jordbruksverket-har-kartlagt-potential-for-inhemsk-godselproduktion (Accessed: September 19, 2025).

RISE (2023) “Svensk elektronik- en ledare i den gröna omställningen.” RISE Research Institutes of Sweden.

RI.SE (2025) *Jordbruksverket har presenterat nya åtgärdsförslag för att stärka den inhemska gödselproduktionen och minska Sveriges importberoende av mineralgödsel. Dessa förslag har nu lämnats till regeringen.*, *Jordbruksverket har presenterat nya åtgärdsförslag för att stärka den inhemska gödselproduktionen och minska Sveriges importberoende av mineralgödsel. Dessa*

förslag har nu lämnats till regeringen. Available at: https://www.ri.se/sv/jordbruksverket-foreslar-atgarder-for-att-starka-sveriges-godselproduktion (Accessed: September 19, 2025).

Ristic, D. (2008) "Liquified Natural Gas—Overview of Prospectives," *Strategic Planning for Energy and the Environment*, 27(4), pp. 66–72. Available at: https://doi.org/10.1080/10485230809509413.

Rodriguez, W. (2023) *Techno-economic and environmental analysis of alternative pathways for urea fertilizer production*. LiU, Linköpings University.

Sauve, G. *et al.* (2023) "Integrated early-stage environmental and economic assessment of emerging technologies and its applicability to the case of plasma gasification," *Journal of Cleaner Production*, 382, p. 134684. Available at: https://doi.org/10.1016/j.jclepro.2022.134684.

SCB, Statistiska centralbyrån (2025) "Försäljning av mineralgödsel för jord- och trädgårdsbruk 2023/24." SCB. Available at: https://www.scb.se/contentassets/15bf6a7d12c54bdb9d0f7c5cba13be88/mi1002_2023b24_sm_mi30br2501.pdf (Accessed: September 22, 2025).

Schmidt, A., Götze, U. and Sygulla, R. (2015) "Extending the scope of Material Flow Cost Accounting – methodical refinements and use case," *Journal of Cleaner Production*, 108, pp. 1320–1332. Available at: https://doi.org/10.1016/j.jclepro.2014.10.039.

Soliman, M.A. (2019) "Empirical Modeling of Urea Synthesis Chemical Equilibria," *International Journal of Innovative Technology and Exploring Engineering*, 8(12), pp. 727–729. Available at: https://doi.org/10.35940/ijitee.L3024.1081219.

Statistics Sweden (2026) *Electricity generation by type of production 2025*, *SCB.se*. SCB. Available at: https://www.scb.se/en/finding-statistics/statistics-by-subject-area/energy/energy-supply-and-use/monthly-electricity-statistics-including-switches-of-electricity-supplier/pong/tables-and-graphs/electricity-generation-by-type-of-production-2025/ (Accessed: January 20, 2026).

Syakdani, A. *et al.* (2019) "Analysis of Cooler Performance in Air Supply Feed for Nitrogen Production Process Using Pressure Swing Adsorption (PSA) Method," *Journal of Physics: Conference Series*, 1167, p. 012055. Available at: https://doi.org/10.1088/1742-6596/1167/1/012055.

Temegne, N.C. *et al.* (2021) "Agroecology for Agricultural Soil Management," in M.K. Jhariya et al. (eds.) *Sustainable Intensification for Agroecosystem Services and Management*. Singapore: Springer Singapore, pp. 267–321. Available at: https://doi.org/10.1007/978-981-16-3207-5_9.

Ulvestad, M. and Overland, I. (2012) "Natural gas and CO_2 price variation: impact on the relative cost-efficiency of LNG and pipelines," *International Journal of Environmental Studies*, 69(3), pp. 407–426. Available at: https://doi.org/10.1080/00207233.2012.677581.

Valente, A. *et al.* (2020) "Using harmonised life-cycle indicators to explore the role of hydrogen in the environmental performance of fuel cell electric vehicles," *International Journal of*

Hydrogen Energy, 45(47), pp. 25758–25765. Available at: https://doi.org/10.1016/j.ijhydene.2019.09.059.

Vätgas Sverige (2026) *Hydrogen in the industry*, *vatgas.se*. Available at: https://vatgas.se/en/fakta/hydrogen-in-the-industry/ (Accessed: January 4, 2026).

Worldometer (2025) *Current world population*, *Current world population*. Available at: https://www.worldometers.info/world-population/ (Accessed: September 11, 2025).

Wulf, S., Ansari, S.M. and Nyhuis, P. (2013) "Material- und Energieeffizienz in der Produktion: Kontinuierliche Verbesserung der Ressourceneffizienz durch gezielte Mitarbeiterqualifizierung," *ZWF Zeitschrift für wirtschaftlichen Fabrikbetrieb*, 108(10), pp. 778–782. Available at: https://doi.org/10.3139/104.111018.

8. Appendix

This chapter includes: Appendix 1, the impact assessment database from Ecoinvent3 and a table detailing acronyms; Appendix 2, which provides inputs and outputs regarding what mass flows are being calculated; and Appendix 3, which contains calculations for the compression and heating of natural gas and water from room temperature and atmospheric pressure to 27 bar and 900 ºC.

8.1 Appendix 1, Impact assessment database

Acronyms used in this appendix are defined in the first Table 5 below. Table 6 presents the assessment data applied in the model to evaluate the environmental impact of energy and mass flows. The reasoning for selecting these assessment values is provided in the final Table 7.

Table 5. Acronyms for identifying the location in the next table

SimaPro acronyms	Definition
{SE}	Sweden
{Europe without Switzerland}, {RER} & {RoE}	Rest of Europe, not including CH
{CH}	Switzerland
{RoW}	Rest of the world, not including CH
{NO}	Norway
{US}	United States
{DZ}	Algeria
{RU}	Russia

Assessment data from SimaPro, using the GWP 100 method, with uptake in kg CO2 eq and cumulative energy in MJ, converted to kWh. The database is Ecoinvent3, system type "Unit".

Table 6. The factor categorizations are presented in the first column. The wide rows indicate flow type, along with the second-to-last column below them. The wide row includes a region indication acronym, indicated in the table as {}. Table 5 describes the acronyms. The first row under the flow-type presents the data used to measure the Global warming potential (GWP), which is measured in kilograms (kg) of carbon dioxide per kilowatt-hour (kWh) or per kg. The two rows below show similar data on primary energy usage, with measurements of primary non-renewable (PNR) and primary renewable (PR) energy, expressed in kWh per kWh or per kg. The last ID column assigns a number to a flow, which has the choice of flow-type explained in Table 7

Source	Value	Unit	As in model	ID
Swedish {SE}				
Electricity, high voltage {SE}| market for electricity, high voltage				
GWP	3.42E-02	kg_CO_2_eq/kWh	Electricity input	1
PnR energy	1.18E+00	kWh/kWh		
PR energy	8.00E-01	kWh/kWh		
Heat, central or small-scale, natural gas {Europe without Switzerland} | market for heat, central or small-scale, natural gas				
GWP	2.80E-01	kg_CO_2_eq/kWh	Heat input	2

PnR energy	1.35E+00	kWh/kWh		
PR energy	1.27E-02	kWh/kWh		
Heat, for reuse in municipal waste incineration only {SE}| market for heat, for reuse in municipal waste incineration only				
GWP	-2.02E-01	kg_CO_2_eq/kWh	Heat output*	3
PnR energy	-4.74E-01	kWh/kWh		
PR energy	-3.65E-01	kWh/kWh		
Tap water {Europe without Switzerland} | market for water, tap water				
GWP	3.24E-03	kg_CO_2_eq/kg	Water input	4
PnR energy	1.51E-03	kWh/kg		
PR energy	1.31E-04	kWh/kg		
Water, deionized {Europe without Switzerland} | market for water, deionized				
GWP	3.15E-04	kg_CO_2_eq/kg	Water input	5
PnR energy	1.28E-03	kWh/kg		
PR energy	1.23E-04	kWh/kg		
Wastewater, unpolluted {CH}| treatment of wastewater, unpolluted, wastewater treatment				
GWP	8.06E-05	kg_CO_2_eq/kg	Water output	6
PnR energy	1.60E-04	kWh/kg		
PR energy	7.75E-05	kWh/kg		
Carbon dioxide, liquid {RER}| carbon dioxide production, liquid				
GWP	7.31E-01	kg_CO_2_eq/kg	CO_2 input	7
PnR energy	2.22E+00	kWh/kg		
PR energy	2.46E-01	kWh/kg		
CO_2 output (assumed)				
	1.00E+00	kg_CO_2_eq/kg	CO_2 output	8
PnR energy	0.00E+00	kWh/kg		
PR energy	0.00E+00	kWh/kg		
**Average between Natural gas, high pressure {RoE} natural gas, high pressure, import from{NO,DE,NL and RU}|				
GWP	3.67E-01	kg_CO_2_eq/kg	Natural gas input	9
PnR energy	9.28E+01	kWh/kg		
PR energy	2.78E-02	kWh/kg		
European {RoE} or {ReR}	**Same flow values as Sweden if otherwise not stated below.**			
Electricity, high voltage {RER}| market group for electricity, high voltage				
GWP	3.48E-01	kg_CO_2_eq/kWh	Electricity input	10
PnR energy	2.19E+00	kWh/kWh		
PR energy	5.02E-01	kWh/kWh		
50% Renewable {SE}	**Average values between Swedish conditions and those below**			
100% Renewable {SE}	**Same flow values as Sweden if otherwise not stated below.**			
Electricity, high voltage {SE}| electricity production, hydro, run-of-river				
GWP	4.47E-03	kg_CO_2_eq/kWh	Electricity input	11
PnR energy	1.14E-02	kWh/kWh		

PR energy	1.05E+00	kWh/kWh		
Electricity, high voltage {SE}| electricity production, hydro, run-of-river				
GWP	4.47E-03	kg_CO_2_eq/kWh	Heat input	12
PnR energy	1.14E-02	kWh/kWh		
PR energy	1.05E+00	kWh/kWh		
Biogas {CH}| market for biogas				
GWP	1.37E+00	kg_CO_2_eq/kg	Natural gas input**	13
PnR energy	7.21E-01	kWh/kg		
PR energy	2.72E-01	kWh/kg		
1 tkm Transport, pipeline, long distance, natural gas {RER w/o DE+NL+RU}| transport, pipeline, long distance, natural gas				
GWP	7.07E-02	kg_CO_2_eq/kg	Natural gas input**	14
PnR energy	3.03E-01	kWh/kg		
PR energy	1.19E-03	kWh/kg		
50% Renewable, {RoE}	**Average values between European conditions and those below**			
100% Renewable, {RoE}	**Same flow values as Europe if otherwise not stated below**			
Electricity, high voltage {RoW}| electricity production, hydro, run-of-river				
GWP	4.71E-03	kg_CO_2_eq/kWh	Electricity input***	15
PnR energy	1.21E-02	kWh/kWh		
PR energy	1.05E+00	kWh/kWh		
Electricity, high voltage {RoW}| electricity production, hydro, run-of-river				
GWP	4.71E-03	kg_CO_2_eq/kWh	Heat input	16
PnR energy	5.33E-02	kWh/kWh		
PR energy	1.05E+00	kWh/kWh		
Biogas {CH}| market for biogas				
GWP	1.37E+00	kg_CO_2_eq/kg	Natural gas input	13
PnR energy	7.21E-01	kWh/kg		
PR energy	2.72E-01	kWh/kg		
1 tkm Transport, pipeline, long distance, natural gas {RER w/o DE+NL+RU}| transport, pipeline, long distance, natural gas				
GWP	7.07E-02	kg_CO_2_eq/kg	Natural gas input**	14
PnR energy	3.02E-01	kWh/kg		
PR energy	1.19E-03	kWh/kg		

Reasoning and scope extension

Table 7. Argues for why certain flow types were chosen in Table 6. The ID column connects flow types to their arguments in the second column.

ID	Argument
1	Wanted, a generalized electricity mix for Sweden
2	No Swedish version, the best of those available, from natural gas, since that is what was used in the articles
3	*Only used in full IS, if applicable. A simple way to give low-value heat residues
4	No Swedish option, the best options available, should not be a big difference

5	No Swedish option, the best options available, should not be a big difference
6	It should be of fair value. The water has not gone through toxic or greasy substances, {CH} is assumed to be closer in value than {RoW}
7	No Swedish version, and this may be quite different.
8	1kg CO_2 is by definition equal 1kg CO_2 eq emission impact.
9	**Top four suppliers of NG to Europe {NO,US,DZ,Ru}, calculated by amount* impact per supplier and then divided by amount. This to get the general impact, no pure {RoE} was available. See last paragraph of scope for more information.
10	Wanted, a generalized electricity mix for Europe
11	Wanted, the lowest emissions renewable electricity option, chosen among several renewable options
12	Wanted: the lowest emissions renewable electricity option, chosen among several renewable options
13	Best of adequate options, better than {RoW}
14	Good option for estimation of pipeline emission per distance, is used with the assumption of Biogas leakage, which probably differ, but its was felt as important to have distance affect biogas in some measure. The value is also used in the estimation of methane impact in Section 5.2.2, where it fits much better
15	*** Lowest impact chosen by normal renewable option, {ReR} was not there, but these values do not alter significantly with geographical location, as seen Sweden's river run off value, it is barely a difference
16	Same reasoning as 12 but for an electricity mix closer to Europe, see reasoning 15.

8.2 Appendix 2, Inputs and outputs within the system barrier

Several outputs are presented in the left column and the corresponding inputs in the second column.

Table 8. General inputs applied to most calculations, molar mass data (Bargiacchi et al., 2021; myengineeringtools, 2025).

Material	Chemical formula	g/mol
Urea	$CO(NH_2)_2$	60.06
Ammonium carbamate	NH_2COONH_4	78.07
Hydrogen gas	H_2	2.016
Nitrogen gas	N_2	28.014
Ammonia	NH_3	17.031
Carbon dioxide	CO_2	44.01
Methane	CH_4	16.04
Water	H_2O	18.015

Table 9. Calculated inputs and outputs for H_2 production using the Methane steam reforming production technology

Outputs	**Inputs**
H_2	Methane
Waste water	Water
CO_2	Electricity
Heat	Heat

Table 10. Calculated inputs and outputs for H_2 production using water electrolysis production technology

Outputs	**Inputs**
H_2	Deionized water

O2	Electricity

Calculated inputs and outputs of N_2 production, Pressure swing adsorption

Outputs	Inputs
N_2	Electricity

Table 11. Calculated inputs and outputs for ammonia synthesis using the Haber-Bosch production technology

Outputs	Inputs
Ammonia	H_2
Heat	N_2
	Electricity
	Heat

Table 12. Calculated inputs and outputs for urea synthesis using the Bosch-Meiser production technology

Outputs	Inputs
Urea	Ammonia
CO_2	CO_2
Heat	Electricity
	Heat

8.3 Appendix 3. Steam methane reforming pre-boundary impacts

The appendix presents the calculations for the heating and compression of water and methane from 25 ºC to 900 ºC and from 1 bar to 27 bar.

8.3.1 Heating data from water and methane at 1 bar.

The following table lists the specific heat capacity under constant pressure for methane, liquid water, and steam at various temperatures under 1 bar. It also details the energy required to boil saturated water under 1 bar.

Table 13. Details specific heat capacity under constant pressure for methane, liquid water, and steam at various temperatures under 1 bar. The temperatures are directly under the material names, and the corresponding specific heat capacity is to the right of those columns. The exception is the specific energy of water evaporation at 1 bar, which is defined for the evaporation process at that pressure (Engineering ToolBox, 2025c, 2025e, 2025f).

Methane	Cp	Water	Cp	Evaporation of water		Steam	Cp
K	kJ/kg*K	*C	kWh/kg*K	*C	kWh/kg	K	kJ/kg*K
300	2.226	25	0.001149	100	0.7433333	300	1.864
350	2.365	50	0.001118			400	1.901
400	2.525	70	0.00109			500	1.954
450	2.703	100	0.001047			600	2.015
500	2.889					700	2.08
550	3.074					800	2.147
600	3.256					900	2.217

650	3.432	1000	2.288
700	3.602	1100	2.358
750	3.766		
800	3.923		
850	4.072		
900	4.214		
950	4.348		
1000	4.475		
1050	4.595		

The average specific heat values between temperatures were estimated below, without using interpolation, and based on the table above

Table 14. Average specific heat capacity for methane, water, and water vapor. The specific heat for evaporation is in the bottom row (Engineering ToolBox, 2025c, 2025e, 2025f).

Specific heat Cp	**Average between 25-900 ºC**	**Column1**
Methane	3.467	kJ/kg*K
Methane	0.000963	kWh/kg*K
Liquid water	0.00110	kWh/kg*K
Water vapor	2.12	kJ/kg*K
Water vapor	0.000589	kWh/kg*K
Evaporation	0.627	kWh/kg

Table 15 is the sum of Table 14 and the incorporation of data from Chisalita, *et al.* (2020).

Table 15. The energy requirement necessary to heat the material specified in the leftmost column from 25 ºC to 900 ºC per kilogram of its own weight. The rightmost column adjusts the kg so that the energy requirement to heat the material necessary to produce and capture 1 kg of H_2. and then the heat requirement to heat the same material (Chisalita, et al. 2020; Engineering ToolBox, 2025b, 2025a, 2025c).

Energy demand input to model	**kWh/kg**	**kg/kg H_2**
Steam	1.18	8.89
Methane	0.842	2.69

8.3.2 Compressor data from water and methane at 900 ºC

Several assumptions were needed to perform the compression, starting with Equation 1 for compressing gases.

Equation 1. Energy equation for the compressor when compressing gases, with efficiency as a factor (Abdoos, Di Benedetto and Russo, 2025)

$$W = \frac{\gamma}{\gamma - 1} \cdot \frac{RT}{\eta} \cdot \left[\left(\frac{P_{out}}{P_{in}} \right)^{\frac{\gamma}{\gamma - 1}} - 1 \right] \text{kJ/kg}$$

With the following other assumptions in Table 16

Table 16. The compressor column lists the symbols used in Equation 1 and the values assigned to the equation. Note that Equation 1 is used two times, one time for water vapor and one time for methane (Abdoos, Di Benedetto and Russo, 2025; Engineering ToolBox, 2025a, 2025b, 2025c, 2025f)

Compressor	Value	Unit	Comment
T	1173	K	Assumed temperature
Γ	1.33	Cp/Cv	for water vapor
Γ	1.32	Cp/Cv	for methane
R	0.462	kJ/kg·K	specific gas constant for water
R	0.518	kJ/kg·K	specific gas constant for methane
η	0.85		assumed compressor efficiency

The energy demand for compression per kg is shown below and is implemented in the model. Note that the energy needed is electricity.

Table 17. The energy requirement necessary to compress the gas specified in the leftmost column from 25 ºC to 900 ºC per kilogram of its own weight. The rightmost column adjusts the kg so that the energy requirement to heat the material necessary to produce and capture 1 kg of H_2. and then the heat requirement to heat the same material (Chisalita et al., 2020).

Material	kWh/kg	kg/kg H_2
Water vapor	0.99085	8.88721
Methane	1.079739	2.689222